Refined BRD - DOPAMS Implementation Specification

Implementation-ready, RFP-traceable, and AI-development ready

Prepared From:

- Source A: Tender Notice TGTSL-ADCS/CS/Eagle Force/2026-DOPMAS dated 28 Feb 2026 (Tender - DOPAMS.pdf)
- Source B: BRD - DOPAMS (BRD - DOPAMS.pdf)

Document Purpose: Transform the supplied BRD into an implementation-ready specification that is traceable to the tender/RFP, measurable, testable, and usable by backend, frontend, API, database, workflow-engine, and AI-assisted development teams.

Traceability Convention: Stable RFP clause IDs have been assigned in this document because the tender text is narrative and partially tabular. These IDs are internal to this refined BRD and are used only for requirement traceability.

1. Executive Summary

1.1 Business Objective

DOPAMS shall be an on-prem AI intelligence platform for EAGLE Force Telangana that consolidates criminal, telecom, financial, legal, grievance, and monitored digital-content data into a single operational system for narcotics intelligence, investigation support, document generation, alerts, and auditability. The platform shall replace fragmented manual workflows with a controlled, evidence-backed, and human-review-driven operating model.

1.2 System Goals

- Build a canonical subject intelligence record using the Department-approved 54-column business schema plus normalized supporting entities.
- Ingest structured and unstructured data from all approved Phase 1 sources and preserve originals with chain-of-custody controls.
- Generate governed outputs such as interrogation reports, technical analysis reports, memos, requisitions, and MIS reports with full source lineage.
- Provide search, dossier, network analysis, geofence alerts, lead management, and risk-based prioritization.
- Ensure every AI-assisted output is explainable, reviewable, and produced only on Department-controlled infrastructure.

1.3 Expected ROI / Target Operating Outcomes

- Reduce manual subject-history collation effort by 60% to 80%, subject to baseline validation during inception.
- Reduce initial interrogation report draft preparation from hours to minutes for data-rich cases.
- Reduce Monthly Report consolidation from multi-day manual compilation to same-day automated aggregation with exception-based review.
- Improve legal-status freshness through scheduled court monitoring and traceable update workflows.
- Achieve 100% digital audit trail coverage for evidence access, approval actions, exports, and lead routing.

1.4 Strategic Alignment

- Supports statewide narcotics intelligence modernization and cross-district visibility.

- Preserves data sovereignty through on-prem deployment and open-source model use only.
- Standardizes legally sensitive outputs through fixed templates, named approvals, and evidence citation.
- Creates a platform foundation for future optional connectors and AI-assisted analysis under Department control.

2. Scope Definition

2.1 In Scope

Scope ID	In-Scope Item
SCP-IS-001	Provide an on-prem responsive web platform for DOPAMS on Department-controlled infrastructure.
SCP-IS-002	Implement role-based and jurisdiction-aware access control with named-user approvals and immutable audit logging.
SCP-IS-003	Implement source connectors and ingestion pipelines for all Phase 1 sources listed in the RFP.
SCP-IS-004	Implement OCR, NLP extraction, bilingual review workflow, and provenance storage for Telugu and English documents.
SCP-IS-005	Implement the canonical 54-column subject profile and related case/evidence entities.
SCP-IS-006	Implement Monthly Report ingestion, 20-KPI extraction, central aggregation, dashboards, and exports.
SCP-IS-007	Implement E-Courts monitoring and legal status update workflow.
SCP-IS-008	Implement financial intelligence cross-check rules and Unocross draft generation with approval controls.
SCP-IS-009	Implement fixed-template interrogation report generation with manual completion workflow.
SCP-IS-010	Implement search, transliteration-aware matching, dossier generation, and subject deduplication/merge controls.
SCP-IS-011	Implement deep link analysis, tower dump processing, technical analysis reports, and geofence/watchlist alerts.
SCP-IS-012	Implement AI-driven content categorization, risk scoring, and legal section mapping with human review.
SCP-IS-013	Implement grievance intake, memo routing, alerting, escalation, and workflow SLA management.
SCP-IS-014	Implement reporting, MIS, scheduled exports, and admin-configurable templates/rules/master data.
SCP-IS-015	Implement model governance, validation tracking, and approved on-prem LLM inference only.
SCP-IS-016	Implement optional connector framework for Phase 3 systems, disabled by default until approved.
SCP-IS-017	Deliver deployment artifacts, environment prerequisites, training materials, support runbooks, and handover pack.

2.2 Out of Scope

Scope ID	Out-of-Scope Item
SCP-OOS-001	Public cloud hosting of application data, model inference, or evidence storage.
SCP-OOS-002	Use of public LLM APIs, public inference endpoints, or third-party hosted foundation models.
SCP-OOS-003	Citizen-facing public portal or public mobile application in initial release.
SCP-OOS-004	Autonomous legal classification, legal section finalization, or enforcement action without named human approval.
SCP-OOS-005	Automated external transmission to financial/legal authorities without approver sign-off.
SCP-OOS-006	Modification or replacement of source systems of record such as CCTNS, C-DAT, or E-Courts.
SCP-OOS-007	Procurement of hardware or network appliances unless separately approved.
SCP-OOS-008	Phase 3 live connector activation before legal approval, credentials, and source availability are confirmed.
SCP-OOS-009	Biometric face search unless TS-COP/CCTNS integration and legal approval are explicitly granted.
SCP-OOS-010	Covert collection from non-approved private accounts, devices, or channels; only lawfully obtained data is in scope.

2.3 Assumptions

Assumption ID	Assumption
ASM-001	Department will provide lawful access, credentials, network whitelisting, and sample data for each approved source system.
ASM-002	Department will confirm the final 54-column dictionary and the 20 KPI dictionary before UAT sign-off.
ASM-003	Department will approve final templates for interrogation reports, lead memos, and requisition drafts.
ASM-004	Department will nominate business owners for legal review, data quality review, and model approval.
ASM-005	Google Drive may be replaced by an approved equivalent storage bridge if required by Department policy.
ASM-006	OCR quality depends on source scan quality; low-confidence outputs will be routed to manual review.
ASM-007	Internet egress, if needed for E-Courts or approved cloud storage sync, will be explicitly allow-listed.
ASM-008	Biometric search will remain feature-flagged until legal approval and connector readiness are confirmed.
ASM-009	Cross-platform monitoring will only use public or lawfully obtained content and approved departmental sources.
ASM-010	Initial model training data will include the Department-provided 50 ideal FIRs and approved benchmark datasets.
ASM-011	Department will provide geofence coordinates, watchlist governance rules, and target classification criteria.
ASM-012	Environment readiness (servers, OS, DB, object storage, search, network) is a prerequisite for installation.

2.4 Constraints

Constraint ID	Constraint
CNS-REG-001	All data processing, storage, model inference, and model training must occur on Department-controlled infrastructure in India.
CNS-REG-002	Sensitive fields such as Aadhaar, PAN, bank details, passport/visa details, and chat evidence must be masked for unauthorized roles.
CNS-REG-003	Every external draft, memo, or legal section suggestion must undergo named human approval before final use.
CNS-INF-001	Reference environments shall include Development, Test/UAT, and Production at minimum.
CNS-INF-002	System shall support PostgreSQL, object storage, search index, workflow workers, and optional GPU inference nodes.
CNS-INF-003	All integrations shall tolerate intermittent source downtime and recover without corrupting data.
CNS-SEC-001	All APIs and browser sessions shall use TLS 1.2 or higher.
CNS-SEC-002	At-rest encryption shall use AES-256 or Department-approved equivalent for DB, object storage, and backups.
CNS-SEC-003	Application shall maintain immutable audit logs for create/update/approve/export/dispatch actions.
CNS-PER-001	Core user flows shall meet the performance baselines defined in Section 8.
CNS-DEL-001	The solution must satisfy the contractual delivery baseline of 6 weeks from purchase order plus 1 week installation, subject to a documented dependency plan.
CNS-DEL-002	No public APIs or outbound AI service calls are permitted.

3. Stakeholders & Roles

3.1 Permission Level Definitions

Permission Level	Label	Definition
PL-0	Read only	May view permitted records and dashboards within jurisdiction; cannot create, edit, approve, export sensitive artifacts, or administer

Permission Level	Label	Definition
		configuration.
PL-1	Operator	May create/update intake records, upload documents, review extractions, and save drafts within jurisdiction.
PL-2	Analyst	Includes PL-1 plus advanced search, dossier, graph, and report generation over permitted data.
PL-3	Approver	Includes PL-2 plus approve/reject/finalize/dispatch actions for governed outputs.
PL-4	Business Admin	Includes PL-3 plus manage templates, routing rules, master data, KPI dictionaries, and feature flags.
PL-5	Platform Admin	Includes PL-4 plus user provisioning, security settings, infra health, connector credentials, backups, and system maintenance.

3.2 Role Catalogue

Role ID	Role Name	Description	System Permissions Level
ROL-001	Department Sponsor / Program Owner	Owns budget, priorities, acceptance, and executive governance.	PL-3
ROL-002	HQ Administrator	Manages statewide configuration, report definitions, templates, routing, and jurisdiction rules.	PL-4
ROL-003	District Operator	Uploads documents, validates extracted data, completes missing fields, and raises leads.	PL-1
ROL-004	Investigating Officer	Uses subject profiles, interrogation reports, technical reports, and linked evidence for investigation.	PL-2
ROL-005	Supervisory Officer / SP / Zonal Officer	Approves memos, requisitions, dispatches, escalations, and district actions.	PL-3
ROL-006	Intelligence Analyst	Performs search, risk review, link analysis, role classification review, and trend analysis.	PL-2
ROL-007	Legal Reviewer	Validates legal section mapping, legal status updates, and final legal document consistency.	PL-3
ROL-008	Toll-Free / WhatsApp Operator	Captures grievance and lead intake through controlled data-entry screens.	PL-1
ROL-009	Read-Only Auditor	Reviews logs, reports, actions, and evidence access history without edit capability.	PL-0
ROL-010	Department IT / Security Team	Approves hosting, security controls, network boundaries, secrets handling, and log access.	PL-5
ROL-011	System Administrator / DevOps	Maintains on-prem infrastructure, application health, backups, and releases.	PL-5
ROL-012	Integration Service Account	Non-human identity used for connector jobs and service-to-service integration.	PL-5
ROL-013	Model Governance Reviewer	Approves model versions, validation metrics, and production promotion of AI artifacts.	PL-3
ROL-014	Bidder Support Engineer	Provides implementation, hypercare, maintenance, and issue resolution under controlled access.	PL-2

4. Functional Requirements

Each functional requirement below is written as a testable implementation contract. All governed outputs are human-review-driven unless explicitly stated otherwise.

FR-01 - Identity, access control, approvals, and audit

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-TECH-07, RFP-BIZ-07, RFP-TECH-01
Description	The platform shall enforce named-user authentication, role-based and jurisdiction-aware authorization, approval controls for governed outputs, and immutable audit logging across UI, API, export, and dispatch actions.
User Stories	• As an authorized Department user, I want access limited to my role and jurisdiction so that sensitive records are visible only to the correct officers. • As a supervisory officer, I want named-user approval before sensitive outputs are finalized so that accountability is preserved.
Acceptance Criteria (Testable)	• AC-01: System shall support at minimum the roles defined in Section 3 and enforce permissions server-side for UI actions, APIs, exports, downloads, and searches. • AC-02: Every user shall have a jurisdiction scope of STATE, ZONE, DISTRICT, and optionally POLICE_STATION; record visibility shall be filtered accordingly unless an explicit override permission exists. • AC-03: Approval-required artifacts (memo, requisition, final interrogation report, legal section mapping finalization, template activation, model promotion) shall require a named approver and digital audit entry before final status change. • AC-04: Sessions shall support Department SSO/LDAP/AD where available, configurable inactivity timeout, configurable password policy for local fallback accounts, and optional MFA. • AC-05: Audit logs shall capture actor, role, timestamp, source IP/device where available, action, target entity, before/after snapshot for mutable fields, and correlation ID.
Business Rules	• BR-01: Authorization failures shall be denied by default and logged as security events. • BR-02: Shared operational accounts shall be prohibited for approval actions. • BR-03: Sensitive field export requires both permission and a business justification code.
Edge Cases	• Suspended users attempting to continue with an active session must be forced to re-authenticate and be denied further access. • Cross-jurisdiction search may show redacted stubs if the user has intelligence lookup permission but not full-record permission.
Failure Handling	• If the identity provider is unavailable, only break-glass local admin accounts approved by the Department may log in; all such access must be specially logged. • If audit-log persistence is unavailable, the platform shall block approval/finalization actions until logging is restored.

FR-02 - Source connectors and ingestion orchestration

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-P1-03, RFP-P1-04, RFP-TECH-01
Description	The platform shall ingest data and documents from approved source systems through scheduled, manual, or event-driven connector pipelines, while preserving source provenance and processing history.
User Stories	• As the platform, I want to ingest records from multiple sources and formats so that downstream

Field	Detail
	intelligence modules can operate without manual re-entry.
Acceptance Criteria (Testable)	• AC-01: System shall support approved ingestion from CCTNS, C-DAT, CDR, IPDR, FIR repositories, Confession cum Seizure Memos, IR repositories, C-Trace, eSakshya, and Google Drive or an approved equivalent storage bridge. • AC-02: Supported inbound payloads shall include PDF, JPG/JPEG, PNG, TIFF, DOCX, XLSX, CSV, JSON, XML, and ZIP archives with checksum validation. • AC-03: Each ingestion job shall persist job_id, source_system, source_record_key, checksum, received_at, processing_started_at, completed_at, retry_count, status, error_code, and correlation_id. • AC-04: Connector retries shall default to 3 attempts with exponential backoff of 5 minutes, 15 minutes, and 60 minutes, after which the job shall move to a failure queue for administrator review. • AC-05: Original documents and raw payloads shall be written to immutable evidence storage before transformation.
Business Rules	• BR-01: A duplicate raw payload identified by identical source_system plus checksum plus source_record_key shall not be processed twice unless a manual reprocess flag is set. • BR-02: Connector-specific credentials shall be stored only in approved secrets storage and never in business tables. • BR-03: Optional Phase 3 connectors shall remain disabled by default via feature flags.
Edge Cases	• Large files exceeding configured limits shall be quarantined with a user-visible validation error and admin alert. • Partial source payloads missing mandatory metadata shall be stored but marked INCOMPLETE_SOURCE for review.
Failure Handling	• A failed connector shall not block unrelated connectors or manual uploads. • When object storage write fails, the job shall terminate in FAILED state and no derived processing shall begin.

FR-03 - OCR, extraction, and bilingual review workflow

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-P1-13, RFP-P1-11, RFP-TECH-02
Description	The platform shall extract structured data from Telugu and English documents using OCR/NLP pipelines and route uncertain outputs to human review with field-level provenance.
User Stories	• As a district operator, I want the system to extract fields from FIRs, Panchanamas, memos, and supporting documents so that manual data entry is minimized.
Acceptance Criteria (Testable)	• AC-01: System shall perform Telugu and English OCR on scanned documents and preserve page-level text, image region references, and extraction confidence values. • AC-02: Each extracted field shall store source document ID, page number, bounding box or text span when available, machine value, reviewer value, confidence score, and extraction version. • AC-03: Default confidence handling shall be configurable and seeded as: >=0.90 auto-accept candidate, 0.60-0.89 review required, <0.60 reject/manual entry required. • AC-04: Review UI shall display source document and extracted fields side-by-side and allow reviewer accept/correct/reject actions per field. • AC-05: Manual reviewer edits shall create a new versioned assertion and shall not overwrite prior machine or human values without history.
Business Rules	• BR-01: Mandatory fields for governed templates may not be marked complete unless a reviewer accepts a value or explicitly records NOT_AVAILABLE with reason code. • BR-02: OCR language detection shall default to bilingual mode when the language is uncertain.

Field	Detail
	• BR-03: Extraction pipelines shall use schema-bound outputs only; free-text hallucinated fields are prohibited.
Edge Cases	• Mixed Telugu-English documents shall support field extraction from both scripts in the same document. • Low-resolution image pages shall still be stored as evidence even if OCR fails.
Failure Handling	• If OCR engine fails for a page, the job shall continue for other pages and flag PAGE_OCR_FAILED for review. • If extraction schema validation fails, the output shall be rejected and the job routed to technical review.

FR-04 - Canonical 54-column subject history builder

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-P1-01, RFP-P1-02, RFP-P2-08
Description	The platform shall build and maintain a canonical subject profile by combining structured and unstructured evidence from approved sources into the Department-approved 54-column schema plus supporting relational entities.
User Stories	• As an investigating officer, I want a unified subject profile so that I can understand identity, crime history, legal status, technical links, and intelligence context in one place.
Acceptance Criteria (Testable)	• AC-01: System shall maintain one canonical subject profile per resolved person/entity cluster and persist the 54-column business schema defined in Section 5.1. • AC-02: Every profile field value shall maintain provenance, confidence, source trust ranking, status (AUTO_PROPOSED, REVIEWED, APPROVED, CONFLICTING, NOT_AVAILABLE), and effective timestamp. • AC-03: Duplicate subject resolution shall apply exact-identifier matching first, then fuzzy match scoring on name, alias, phone, address, and relationship signals; auto-link threshold shall be configurable. • AC-04: Profiles shall display completeness score, unresolved conflicts count, latest update timestamp, and profile version history. • AC-05: Subject photos from CCTNS or approved sources shall be stored as evidence references and linked to the profile when permitted.
Business Rules	• BR-01: A lower-confidence or lower-trust source value shall not automatically replace an approved higher-trust value. • BR-02: Conflicting values shall be stored as separate assertions until reviewer resolution. • BR-03: Merge operations shall preserve source lineage for all inherited assertions.
Edge Cases	• Different subjects sharing a phone number or address shall remain separately resolvable when other identifiers differ. • Subjects without exact identifiers shall still be creatable as provisional profiles with a dedupe review flag.
Failure Handling	• If match scoring service is unavailable, new evidence shall be attached to a provisional subject and queued for merge review. • If profile publication fails after approval, the approved version shall remain recoverable from the workflow transaction log.

FR-05 - Monthly Report ingestion and KPI consolidation

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-P1-05, RFP-P1-06, RFP-P1-07, RFP-TECH-06
Description	The platform shall detect district Monthly Reports, extract the approved KPI set, and consolidate final values into central reporting stores with traceable review history.
User Stories	• As HQ staff, I want district monthly reports parsed automatically so that statewide statistics do not require manual aggregation.
Acceptance Criteria (Testable)	• AC-01: System shall poll or receive events from a configured folder/source and detect new or modified MR files by checksum and metadata. • AC-02: System shall extract the Department-approved 20 KPI parameters and persist both machine-parsed and reviewer-approved values. • AC-03: A district plus reporting_month combination shall have at most one FINAL approved KPI record; re-uploads shall create a new version and supersede the prior final only after review. • AC-04: Low-confidence, missing, or format-invalid KPI values shall route the report to REVIEW_REQUIRED state with field-level issue codes. • AC-05: Final approved KPI values shall be available to dashboards, exports, and scheduled MIS reports within 5 minutes of approval.
Business Rules	• BR-01: KPI dictionary entries shall define metric_id, label, data type, allowed range/pattern, aggregation rule, and source note. • BR-02: Reports without district and reporting month metadata shall not finalize. • BR-03: Reviewer corrections shall require reason codes when they differ from machine values.
Edge Cases	• Multi-district combined reports must be rejected unless explicitly allowed by configuration. • Scanned MR documents with embedded tables may require manual confirmation for table extraction.
Failure Handling	• If storage sync fails, the last successful watermark shall be preserved and retry scheduled. • If KPI publication to reporting mart fails, the approved KPI record shall remain stored and an admin alert shall be raised.

FR-06 - E-Courts legal status monitoring

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-P1-08
Description	The platform shall periodically search approved court sources for judgments, bail orders, and legal status changes and update subject/case records through a controlled review workflow.
User Stories	• As a legal reviewer, I want case outcomes and bail orders to update subject records so that legal status stays current.
Acceptance Criteria (Testable)	• AC-01: System shall search E-Courts using configured keys such as case number, accused name, court name, year, and district context. • AC-02: Open or monitored cases shall be re-checked on a configurable schedule, seeded at once every 24 hours. • AC-03: Exact or high-confidence matches above configured threshold shall generate a proposed legal update with order metadata and evidence attachment. • AC-04: Ambiguous matches shall route to manual legal review and shall never auto-apply legal status changes. • AC-05: Prior legal statuses shall remain historically queryable and compare-able after new status

Field	Detail
	application.
Business Rules	• BR-01: Status changes shall store order type, order date, court details, source URL/reference, retrieval timestamp, and parser version. • BR-02: If multiple later orders exist, the newest legally effective order shall be proposed as current while preserving older orders. • BR-03: Monitoring must stop only when case closure rule configured by Department is satisfied.
Edge Cases	• Name-only matches with multiple accused candidates must not auto-link. • Unavailable court pages due to captcha/network controls must be logged as SOURCE_UNAVAILABLE rather than interpreted as no update.
Failure Handling	• If retrieval fails repeatedly beyond threshold, monitored case shall generate a stale-status alert. • If parser fails on a retrieved order, the raw document shall still be stored and assigned for review.

FR-07 - Financial intelligence cross-check and Unocross draft generation

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-P1-09, RFP-P1-10
Description	The platform shall correlate transaction-linked phone/account data with telecom identities, evaluate configurable trigger rules, and generate approval-controlled requisition drafts for additional bank data.
User Stories	• As an intelligence analyst, I want repetitive transaction patterns to trigger a requisition draft so that likely supplier and peddler relationships can be escalated quickly.
Acceptance Criteria (Testable)	• AC-01: System shall normalize transaction identifiers, account references, UPI IDs, and phone numbers before matching against subject, telecom, and device data. • AC-02: Rule engine shall support configurable thresholds for recurrence count, time window, amount, subject/counterparty recurrence, and telecom match confidence. • AC-03: When rule conditions are met, the system shall create a pre-filled Unocross draft containing case references, suspected parties, linked numbers, trigger explanation, and supporting evidence summary. • AC-04: Drafts shall follow workflow states DRAFTED, REVIEW_PENDING, APPROVED, REJECTED, DISPATCHED, EXPIRED and require named approver sign-off before export or send. • AC-05: Approved drafts shall be exportable as PDF and email-ready text and shall preserve the exact trigger snapshot used at approval time.
Business Rules	• BR-01: Partial evidence without sufficient telecom linkage may create an analyst review task but shall not auto-generate a formal draft. • BR-02: Duplicate triggers for the same subject/counterparty pair inside suppression window shall update the existing draft record instead of creating a new one. • BR-03: Trigger rules are configuration data and require admin approval before activation.
Edge Cases	• Transactions involving shared family accounts may create false positives and therefore require evidence explanation and reviewer confirmation. • Bulk imported bank statements spanning multiple months may trigger several candidate pairs in one job.
Failure Handling	• If financial data ingestion succeeds but telecom matching fails, the system shall preserve unmatched evidence and mark MATCH_PENDING. • If PDF generation fails for an approved draft, users shall still be able to retrieve email-ready text and see the failure log.

FR-08 - Fixed-template interrogation report generation

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-P1-11, RFP-P1-12, RFP-P1-13
Description	The platform shall generate interrogation reports using Department-approved templates, populated from primary and supplementary sources, with explicit manual completion and approval controls.
User Stories	• As a district operator, I want the system to populate a mandatory interrogation report template so that consistent legally usable reports can be produced quickly across districts.
Acceptance Criteria (Testable)	• AC-01: System shall ingest FIR as the primary source and supplement with Panchanama, CCTNS, CDR, IPDR, Confession/Seizure Memo, IR, C-Trace, and eSakshya data where available. • AC-02: Template versions shall be centrally managed and only ACTIVE versions may be used for new report generation. • AC-03: Missing mandatory template fields shall be presented in a structured form with validation and status codes before finalization. • AC-04: Final reports shall be exportable as PDF and, where permitted by role, editable DOCX while preserving template version and source lineage metadata. • AC-05: Finalization shall require review completion and named approver sign-off when configured by jurisdiction.
Business Rules	• BR-01: Generated narrative sections must be grounded in cited source evidence and must not invent facts. • BR-02: Template schema changes shall not retroactively alter previously finalized reports. • BR-03: Users without finalization permission may save drafts only.
Edge Cases	• Cases with missing FIR metadata must remain in DRAFT state until minimum case identifiers are supplied. • Multiple supplementary documents with conflicting facts must be flagged for reviewer resolution.
Failure Handling	• If document merge/export fails, the report draft and all populated fields shall remain recoverable. • If required template package is unavailable, report generation shall be blocked with CONFIGURATION_ERROR.

FR-09 - Unified search, transliteration-aware matching, and one-click dossier

Field	Detail
Priority	Should Have
Target Release	R2
RFP Traceability	RFP-P2-01, RFP-P2-08
Description	The platform shall provide a unified search experience across identities, cases, phones, vehicles, and approved biometrics and shall generate a consolidated dossier view for a selected subject.
User Stories	• As an analyst, I want to search by multiple identifiers and get an instant dossier so that I can assess a subject quickly.
Acceptance Criteria (Testable)	• AC-01: Search shall support full name, alias, surname, phone number, crime number, Aadhaar, PAN, vehicle identifier, document ID, and biometric face match where approved integration exists. • AC-02: Search shall support fuzzy matching, transliteration-aware Telugu/English matching, and ranking reasons for each result candidate. • AC-03: One-click dossier shall include identity summary, case history, role/status, legal status, documents, alerts, associates, and latest intelligence indicators where available. • AC-04: Dossier export shall be role-restricted, watermarked, and auditable. • AC-05: Users shall be able to launch link analysis, technical analysis, or report generation directly from the dossier when permitted.

Field	Detail
Business Rules	• BR-01: Exact identifier matches rank above fuzzy matches. • BR-02: Face-match search shall remain hidden when connector or approval flag is disabled. • BR-03: Masked identifiers shall remain masked in search results unless role permits full view.
Edge Cases	• Very common names shall return paginated results with district filters and match explanations. • Aliases with phonetic variations across Telugu and English shall be searchable using transliteration rules.
Failure Handling	• If search index is unavailable, system shall fall back to DB search for basic identifiers with a user-visible performance warning. • If dossier assembly cannot reach one enrichment source, partial dossier shall still render with unavailable-section markers.

FR-10 - Natural-language query and insight assistant

Field	Detail
Priority	Should Have
Target Release	R2
RFP Traceability	RFP-TECH-03, RFP-TECH-02
Description	The platform shall provide an on-prem natural-language assistant for grounded search, summarization, and analytic insight over approved datasets.
User Stories	• As an authorized officer, I want to ask the platform questions in plain language so that I can retrieve intelligence quickly without manually building filters.
Acceptance Criteria (Testable)	• AC-01: Assistant shall accept natural-language questions and translate them into approved retrieval or analytics operations over permitted data only. • AC-02: Every answer shall include cited source references, applied filters or time window, and an explanation of whether the output is summary, count, trend, or recommendation. • AC-03: Assistant shall run only on approved on-prem open-source models and shall not call public inference APIs. • AC-04: Users shall be able to convert an answer into a saved search, report draft, or dashboard filter where applicable. • AC-05: Prompt and response metadata, including model version and token counts, shall be logged for governance without exposing sensitive content to unauthorized roles.
Business Rules	• BR-01: The assistant must refuse actions outside user permissions and respond with a controlled access message. • BR-02: Unsupported facts shall be returned as 'insufficient evidence' rather than hallucinated conclusions. • BR-03: AI-generated recommendations are advisory only and require human interpretation.
Edge Cases	• Queries mixing English and Telugu terms shall still resolve using multilingual retrieval. • Ambiguous date ranges such as 'recent' shall default to the last 30 days unless the user specifies otherwise or saved filters override.
Failure Handling	• If AI service is unavailable, the UI shall surface fallback structured search shortcuts. • If cited sources cannot be produced, the assistant answer shall not be displayed as final.

FR-11 - n-level link analysis and kingpin discovery

Field	Detail
Priority	Should Have
Target Release	R2
RFP Traceability	RFP-P2-02

Field	Detail
Description	The platform shall perform iterative multi-hop link analysis over communication, financial, co-case, and co-location evidence to identify bridge nodes and potential hidden suppliers.
User Stories	• As an intelligence analyst, I want multi-level contact analysis so that hidden supplier or kingpin nodes can be identified beyond direct contacts.
Acceptance Criteria (Testable)	• AC-01: System shall support configurable graph depth from 1 to 5, default 3, and reject requests above the configured cap. • AC-02: Analysis shall rank common nodes, bridge nodes, and high-centrality nodes using configurable scoring factors such as call frequency, duration, cross-seed convergence, and recency. • AC-03: Graph jobs shall run asynchronously and expose status, progress, completion time, and result retention metadata. • AC-04: Graph UI shall explain why each key node is ranked using evidence counts and factor breakdown. • AC-05: Users shall be able to filter by date range, source, duration threshold, jurisdiction, and subject/case context.
Business Rules	• BR-01: Result sets must record the exact data window and source set used for reproducibility. • BR-02: Deleted or legally held evidence shall not be physically removed from historic graph result provenance; only access shall be restricted. • BR-03: Graph exports shall include legend, scoring basis, and watermark.
Edge Cases	• A subject may appear through multiple node types (person, phone, device, account); UI shall present linkage clearly. • Very large graphs may require auto-pruning of low-weight edges for visualization while retaining full exportable tables.
Failure Handling	• If graph store is unavailable, queued jobs shall remain pending and the user shall see a retriable technical error. • If a source subset cannot be reached, results shall display partial-data disclaimer rather than silent omission.

FR-12 - Automated technical analysis report

Field	Detail
Priority	Should Have
Target Release	R2
RFP Traceability	RFP-P2-03
Description	The platform shall generate a technical analysis report for a selected number or subject using CDR/CDAT/LBS data and configured analytics rules.
User Stories	• As an investigating officer, I want a technical analysis report for a suspect number so that movement, contact, and device patterns can be reviewed quickly.
Acceptance Criteria (Testable)	• AC-01: System shall generate PDF reports containing route maps, stay locations, top 10 contacts, duration patterns, IMEI history, and silence or switch-off patterns when source data exists. • AC-02: Home and office inference shall use configurable dwell-time and time-of-day rules and label results as inferred rather than confirmed facts. • AC-03: Every chart, map, and section shall display the source data range, generation timestamp, and report version. • AC-04: Missing data sections shall render 'Data unavailable' explicitly rather than blank content. • AC-05: Users shall be able to regenerate the report for a different time window without altering historic versions.
Business Rules	• BR-01: Report sections derived from inference must carry a confidence label. • BR-02: Top contact ranking criteria shall be configurable and stored with the report version. • BR-03: LBS access must be role-restricted.

Field	Detail
Edge Cases	• Subjects using multiple devices or SIM swaps shall show segmented history rather than one flattened timeline. • Location data with poor accuracy shall be labeled low confidence.
Failure Handling	• If map tile service is unavailable on the intranet, the report shall still generate table-based location summaries. • If one analytic sub-component fails, remaining sections shall render with section-level error notes.

FR-13 - Geo-fencing and watchlist alerts

Field	Detail
Priority	Should Have
Target Release	R2
RFP Traceability	RFP-P2-04, RFP-BIZ-07
Description	The platform shall monitor watchlisted subjects or numbers against configured geofences and create actionable alerts with acknowledgement and escalation controls.
User Stories	• As a supervisory officer, I want geofenced alerts for high-value targets so that field teams can be informed when watchlisted subjects enter sensitive zones.
Acceptance Criteria (Testable)	• AC-01: System shall maintain a watchlist with an initial operational target of 50 high-value subjects and support priority tiers. • AC-02: System shall support predefined geofences for Goa, Bengaluru, Mumbai, and Orissa Agency plus Department-defined custom zones with geospatial polygons. • AC-03: Entry, exit, and dwell events shall generate alerts with subject reference, event time, source, geofence, severity, and recommended action. • AC-04: Duplicate alerts within a configurable suppression window, seeded at 30 minutes, shall be suppressed or merged. • AC-05: Alerts shall support ACKNOWLEDGED, ASSIGNED, ESCALATED, RESOLVED, CLOSED, and FALSE_POSITIVE statuses.
Business Rules	• BR-01: Only approved high-value watchlist items may generate real-time alerting. • BR-02: Severity shall be derived from watchlist priority, geofence sensitivity, and recency of prior alerts. • BR-03: Escalation timers shall be configurable by severity.
Edge Cases	• Repeated oscillation across a zone boundary shall not spam duplicate alerts because suppression or merge logic must apply. • Source location accuracy below configured threshold shall reduce severity or create a review alert instead of direct field escalation.
Failure Handling	• If real-time feed is delayed, the alert shall be marked LATE_EVENT and still stored for audit. • If geofence computation service is unavailable, incoming events shall queue for retry.

FR-14 - Tower dump analytics and rank ordering

Field	Detail
Priority	Should Have
Target Release	R2
RFP Traceability	RFP-P2-05
Description	The platform shall process tower dump datasets, normalize telecom identifiers, and rank candidate numbers by criminal or crime-link strength.
User Stories	• As an analyst, I want tower dump numbers ranked by offender and crime link strength so that priority suspects can be identified faster.

Field	Detail
Acceptance Criteria (Testable)	• AC-01: System shall accept approved bulk tower dump formats and validate file structure before processing. • AC-02: Imported numbers shall be normalized to canonical format and matched against subject, case, device, and prior event stores. • AC-03: Output shall rank results in descending order using configurable criminal-link and crime-link scoring factors and show factor breakdown. • AC-04: Tower dump jobs shall run asynchronously with progress visibility and downloadable result tables. • AC-05: Results shall be retained with source file hash, processing version, and query parameters for audit.
Business Rules	• BR-01: Numbers failing normalization shall be retained in an exception report and excluded from ranking. • BR-02: Ranking formulas are configuration artifacts and require admin approval before changes. • BR-03: Large jobs may chunk processing but final ranking must be deterministic.
Edge Cases	• The same number may appear across multiple source towers and time windows; ranking shall account for recurrence. • International format numbers and missing country codes must normalize consistently.
Failure Handling	• If worker capacity is exceeded, jobs shall queue rather than fail immediately. • If ranking stage fails after normalization, normalized data shall be preserved for reprocessing.

FR-15 - Drug offender role classification and pattern analysis

Field	Detail
Priority	Should Have
Target Release	R2
RFP Traceability	RFP-P2-06, RFP-P2-07, RFP-P2-08, RFP-BIZ-03
Description	The platform shall classify suspects into controlled offender roles using Department-provided training examples, extraction rules, and supporting evidence from multi-source data.
User Stories	• As an analyst, I want the system to classify suspect roles from case facts so that investigators can prioritize actions using consistent patterns.
Acceptance Criteria (Testable)	• AC-01: System shall support roles Cultivator, Manufacturer, Supplier, Peddler, Transporter, Consumer, Mediator, and Financier. • AC-02: Model or rule pipeline shall be tuned using the Department-provided 50 ideal FIRs plus approved extraction rules and benchmarked before promotion. • AC-03: Each classification shall return role label, confidence, supporting evidence snippets, source references, and model/rule version. • AC-04: Low-confidence or conflicting classifications shall route to manual review rather than auto-finalize. • AC-05: Classification shall support natural, synthetic, and semi-synthetic drug contexts under Department-approved NDPS taxonomy.
Business Rules	• BR-01: Classification may recommend multiple candidate roles but only approved final role(s) shall appear as active profile roles. • BR-02: The system must preserve historical role decisions and reasons when updated. • BR-03: Benchmark thresholds are governed by model governance in FR-26.
Edge Cases	• One subject may act as both transporter and peddler across cases; model output must allow multi-role history. • Confession facts may conflict with FIR facts and require reviewer adjudication.
Failure Handling	• If the model version falls below approved benchmark, production must revert to manual-only or rule-only mode.

Field	Detail
	• If training dataset ingestion fails, no model promotion may proceed.

FR-16 - Public grievance and lead management

Field	Detail
Priority	Should Have
Target Release	R2
RFP Traceability	RFP-P2-09, RFP-P2-10, RFP-BIZ-07
Description	The platform shall capture public grievances and operational leads in a controlled interface, enrich them with known context, generate official memos, and route them through approval and action workflows.
User Stories	• As a toll-free or WhatsApp operator, I want to enter complaint details directly into the system so that leads are stored centrally and routed formally.
Acceptance Criteria (Testable)	• AC-01: Lead entry shall capture at minimum informant name, contact/channel, location, suspect details, nature of information, urgency, free-text notes, and optional attachments. • AC-02: On submit, system shall create a permanent lead record with timestamps, source channel, submitting user, and jurisdiction derivation. • AC-03: Duplicate detection shall warn the user when similar leads exist using location, suspect, contact, and recency signals; user may override with justification. • AC-04: System shall generate an official memo using approved template and available DOPAMS context, then route it to the relevant Zonal Officer/SP based on routing rules. • AC-05: Lead statuses shall support NEW, VALIDATED, MEMO_GENERATED, APPROVAL_PENDING, ROUTED, IN_ACTION, CLOSED, and REJECTED.
Business Rules	• BR-01: Submitted leads cannot be hard-deleted by standard users. • BR-02: Routing rules must be configuration-driven and auditable. • BR-03: Critical urgency leads shall trigger immediate alerting under FR-24.
Edge Cases	• Anonymous leads may omit informant name but must carry ANONYMOUS flag and channel metadata. • Leads spanning multiple districts shall support primary and secondary jurisdiction routing.
Failure Handling	• If memo generation fails, the lead shall remain created and route to manual memo preparation queue. • If routing target cannot be resolved, the item shall route to HQ fallback queue.

FR-17 - MIS dashboards and automated reporting

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-P1-07, RFP-TECH-06, RFP-TECH-08
Description	The platform shall provide dashboards, analytics, and scheduled reports over central data for operational, management, and audit use.
User Stories	• As HQ leadership, I want automated dashboards and reports so that Department performance and trends can be monitored without manual compilation.
Acceptance Criteria (Testable)	• AC-01: System shall provide charts, graphs, maps, tabular analytics, and statistical reports over approved subject, case, MR, lead, alert, and workflow datasets. • AC-02: Reports shall support filtering by date range, district, police station, drug type, role, source, status, and watchlist where applicable. • AC-03: Scheduled reports shall support PDF and XLSX exports and delivery to approved internal shared storage or email relay.

Field	Detail
	• AC-04: Simple report layout changes, column order changes, and filter parameter changes shall be configurable by admins without code changes. • AC-05: Dashboards shall display data freshness timestamps and anomaly markers when anomaly rules are enabled.
Business Rules	• BR-01: Reports containing masked fields must preserve masking in exports unless the requesting role has explicit permission. • BR-02: Department-defined MIS templates shall be versioned and approval-controlled. • BR-03: Scheduled distribution lists must use named groups approved by HQ Admin.
Edge Cases	• Empty result sets shall still generate valid reports with 'No data for selected filters'. • Very large XLSX exports shall stream in chunks and provide completion notification.
Failure Handling	• If export generation fails, the job shall be retriable and the user shall see status/error details. • If reporting mart refresh is delayed, dashboards shall show stale-data warning.

FR-18 - Optional external connector framework

Field	Detail
Priority	Nice to Have
Target Release	R3
RFP Traceability	RFP-P1-04
Description	The platform shall provide a pluggable connector framework for optional future systems without redesigning the core data model.
User Stories	• As the Department, I want optional external integrations to be pluggable so that the platform can expand when data access is approved.
Acceptance Criteria (Testable)	• AC-01: System shall support connector registration metadata including auth type, mapping profile, sync mode, and feature flag state. • AC-02: New connector payloads shall map to the canonical entity model through configurable transformation profiles. • AC-03: Connector failures shall not block core application workflows or existing source jobs. • AC-04: Each connector shall be independently enable-able, disable-able, and auditable by admin users. • AC-05: Connector-specific data shall be traceable to its source for review and unmapping if required.
Business Rules	• BR-01: Optional connectors remain disabled until legal approval, credentials, and mappings are signed off. • BR-02: Canonical entity contracts may not be broken by connector-specific custom fields; extensions must use extension tables/json fields with schema governance.
Edge Cases	• A connector may be approved for read-only batch mode but not API mode. • Different state systems may provide overlapping identifiers with inconsistent formats.
Failure Handling	• If a connector schema changes unexpectedly, inbound jobs shall be quarantined rather than silently dropped. • If credentials expire, connector status shall move to DISABLED_WITH_ERROR and notify admins.

FR-19 - Cross-platform monitoring and monitored content ingestion

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-BIZ-02, RFP-BIZ-01
Description	The platform shall ingest approved monitored content from public or lawfully obtained online sources and

Field	Detail
	normalize it as intelligence content items for analysis, categorization, and evidence preservation.
User Stories	• As an intelligence operator, I want monitored online/public content to enter DOPAMS in a structured way so that digital promotion and coordination signals can be reviewed centrally.
Acceptance Criteria (Testable)	• AC-01: System shall support monitored content intake from approved URLs, platform exports, screenshots, documents, or API feeds as permitted by the Department. • AC-02: Each content item shall capture platform, channel or handle, capture time, collector/source, raw content reference, language, media presence, and dedupe key. • AC-03: Content items shall be routed through categorization, risk scoring, and evidence preservation workflows. • AC-04: Duplicate or near-duplicate content shall be clustered using configured text/hash similarity rules. • AC-05: Content visibility shall be role-restricted and follow the same audit controls as other evidence items.
Business Rules	• BR-01: Only approved lawful collection methods may be used; content from prohibited/private sources is out of scope unless separately authorized. • BR-02: Content capture metadata must include who or which connector collected the content and when. • BR-03: Content deletes from source platforms do not remove preserved evidence copies once lawfully ingested.
Edge Cases	• Content may include mixed media and multilingual text requiring separate extraction passes. • The same message may appear in both screenshot and text export form and must cluster as one evidentiary item.
Failure Handling	• If source content cannot be re-fetched later, the preserved captured version shall still remain reviewable. • If media extraction fails, textual metadata shall still be stored and the item marked PARTIAL_MEDIA.

FR-20 - AI-based content categorization, risk scoring, and prioritization

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-BIZ-03, RFP-BIZ-04, RFP-TECH-03
Description	The platform shall classify monitored content, leads, and selected subject events into controlled categories and compute configurable risk scores used for prioritization queues and alerts.
User Stories	• As an analyst, I want content and events prioritized by explainable risk score so that the highest-value items are reviewed first.
Acceptance Criteria (Testable)	• AC-01: System shall assign each monitored content item and eligible lead/event a category from an approved taxonomy including at minimum DRUG_SALE_SIGNAL, COORDINATION_SIGNAL, LOGISTICS_SIGNAL, FINANCIAL_SIGNAL, RECRUITMENT_SIGNAL, BENIGN, and UNKNOWN. • AC-02: System shall calculate a risk score from 0 to 100 using configurable weighted factors such as source credibility, recency, repeated entities, keyword patterns, linked offenders, geofence relevance, and prior alert history. • AC-03: Score responses shall include factor contribution breakdown, model/rule version, confidence, and reviewer override fields. • AC-04: Items above configured thresholds shall enter PRIORITY_REVIEW or CRITICAL_ALERT queues according to workflow rules. • AC-05: Users with review permission shall be able to override category and score with mandatory reason code.
Business Rules	• BR-01: Scoring formulas and weights are governed configuration artifacts and require admin

Field	Detail
	approval before production use. • BR-02: No score may directly trigger an external legal action without human approval. • BR-03: Historical scores shall remain versioned even when configuration changes.
Edge Cases	• A benign content item linked to a high-risk watchlisted subject may still receive elevated priority due to relationship factors. • Items with low model confidence shall still receive category UNKNOWN and route for manual review.
Failure Handling	• If scoring service is unavailable, affected items shall enter UNSCORED_REVIEW queue rather than be silently deprioritized. • If factor data is partially missing, scoring shall proceed only if minimum factor coverage is met; otherwise mark INSUFFICIENT_DATA.

FR-21 - Legal section mapping and reviewer confirmation

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-BIZ-05, RFP-P1-01, RFP-P1-11
Description	The platform shall propose candidate legal sections from extracted facts and entity context, while requiring legal or supervisory review before final application.
User Stories	• As a legal reviewer, I want candidate legal sections suggested from case facts so that drafting is faster but still controlled.
Acceptance Criteria (Testable)	• AC-01: System shall generate candidate legal sections based on extracted facts, offense patterns, controlled vocabularies, and Department-approved legal mapping rules. • AC-02: Each suggestion shall include rationale text, supporting evidence snippets, source references, and confidence or rule strength. • AC-03: Users shall be able to accept, reject, or modify candidate sections and record justification. • AC-04: Final accepted sections shall update case/profile records and governed document drafts with full audit trail. • AC-05: Legal mapping rules shall be versioned, effective-dated, and rollback-capable.
Business Rules	• BR-01: The system may suggest but shall never silently auto-finalize legal sections in official outputs. • BR-02: Section suggestions must cite source evidence; uncited suggestions are invalid. • BR-03: Multiple sections may be active for one case when justified.
Edge Cases	• Facts may support multiple section combinations across NDPS and allied laws. • Conflicting facts from different documents may generate conflicting section suggestions that require review.
Failure Handling	• If legal mapping engine is unavailable, manual legal section entry shall remain possible and auditable. • If the current active legal mapping ruleset is missing, finalization shall be blocked with CONFIGURATION_ERROR.

FR-22 - Digital evidence preservation and chain-of-custody

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-BIZ-06, RFP-P1-03
Description	The platform shall preserve original and derived evidence with integrity controls, legal hold, retention,

Field	Detail
	and complete chain-of-custody history.
User Stories	• As a reviewer or auditor, I want every evidence artifact to retain integrity and access history so that evidentiary trust is maintained.
Acceptance Criteria (Testable)	• AC-01: Every stored original file and every exported governed artifact shall have a SHA-256 hash recorded at creation time. • AC-02: System shall maintain chain-of-custody events for ingest, access, download, export, approval, dispatch, legal hold, restore, and archive actions. • AC-03: Evidence shall support legal hold flagging that prevents purge or archival deletion until released by authorized role. • AC-04: Retrieval API shall support integrity verification by recalculating and comparing stored hash values on demand. • AC-05: Evidence access and export shall be watermarkable, role-restricted, and fully auditable.
Business Rules	• BR-01: Standard users may not permanently delete evidence artifacts. • BR-02: Derived artifacts must reference their parent evidence set and generation parameters. • BR-03: Retention policies shall be configurable by evidence type and legal hold status.
Edge Cases	• The same file ingested through two channels shall preserve both provenance records while sharing binary storage only if hashes match and policy allows dedup. • Evidence attached to multiple cases must retain all case links.
Failure Handling	• If hash verification fails, the artifact shall be quarantined and a critical security alert shall be raised. • If legal hold service is unavailable, purge jobs shall suspend execution.

FR-23 - Template, master data, and rules administration

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-P1-12, RFP-TECH-08, RFP-TECH-06
Description	The platform shall allow authorized administrators to manage templates, KPI dictionaries, routing rules, geofences, legal mapping rules, and scoring weights without code changes for simple configuration updates.
User Stories	• As an HQ administrator, I want to update templates and rules centrally so that districts use the latest approved configuration without software redeployment.
Acceptance Criteria (Testable)	• AC-01: Admin UI shall manage templates, KPI definitions, routing rules, alert thresholds, geofences, risk scoring weights, and master data enumerations. • AC-02: Config changes shall support draft, review, approve, activate, deactivate, and rollback states with effective dates. • AC-03: Each active configuration item shall expose version number, activation timestamp, approver, and change reason. • AC-04: Config updates for simple metadata/rules shall take effect without application redeployment. • AC-05: Historic reports and decisions shall remain linked to the version active at their execution time.
Business Rules	• BR-01: Only PL-4 or higher users may create or activate governed configuration changes. • BR-02: Activation of legal, scoring, or workflow rules requires two-step approval if Department policy enables maker-checker mode. • BR-03: Deleted master data values must remain visible for historic records but unavailable for new records.
Edge Cases	• A template may be active statewide but temporarily overridden for one district only if allowed by policy. • Geofence boundaries may overlap; rule priority must be explicit.

Field	Detail
Failure Handling	• If configuration cache refresh fails, the prior active version shall remain in effect. • If activation results in validation errors, the new version shall not go live.

FR-24 - Notifications, escalation, and SLA management

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-BIZ-07, RFP-P2-10
Description	The platform shall support workflow notifications, reminder timers, escalation rules, and SLA tracking for alerts, reviews, approvals, and lead routing.
User Stories	• As an operations supervisor, I want overdue items escalated automatically so that critical intelligence actions are not missed.
Acceptance Criteria (Testable)	• AC-01: System shall support event-triggered notifications for new alerts, review-required documents, approval tasks, failed connectors, and critical leads. • AC-02: SLA timers shall be configurable by workflow type and severity and shall track due_at, acknowledged_at, escalated_at, and closed_at timestamps. • AC-03: Escalation rules shall support hierarchical escalation to supervisory officers or HQ fallback queues after configurable timeout. • AC-04: Notification channels shall include in-app notifications and Department-approved email relay; additional channels may be enabled by configuration. • AC-05: Users shall be able to acknowledge, assign, snooze within policy, or close tasks with reason codes.
Business Rules	• BR-01: Critical alerts may not be snoozed by operator-level users. • BR-02: Escalation must preserve the original assignee and escalation trail. • BR-03: Closed tasks require closure code and acting user.
Edge Cases	• An item may change severity after enrichment, requiring timer recalculation. • One event may notify multiple roles but only one primary owner should be accountable.
Failure Handling	• If email relay is unavailable, in-app notifications shall still be created and the failure logged. • If scheduler service is delayed, overdue calculations shall catch up on recovery.

FR-25 - Subject deduplication, merge, and survivorship management

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-P1-02, RFP-P2-08
Description	The platform shall manage duplicate subject candidates and provide controlled merge operations with field survivorship, lineage preservation, and conflict auditability.
User Stories	• As a reviewer, I want to merge duplicate subject profiles safely so that intelligence is not fragmented across multiple records.
Acceptance Criteria (Testable)	• AC-01: System shall generate duplicate candidate queues using configurable match rules and thresholds. • AC-02: Merge review UI shall compare candidate profiles side-by-side with field-level provenance and recommended survivorship. • AC-03: Final merge shall produce a single surviving subject_id plus alias mappings and redirect links from merged IDs. • AC-04: Merge operations shall be fully auditable and reversible only by authorized admin through a

Field	Detail
	controlled restoration workflow. • AC-05: Post-merge, all linked cases, documents, alerts, and graph nodes shall resolve to the surviving subject while retaining historic source IDs.
Business Rules	• BR-01: Exact government identifier conflicts require manual review and cannot auto-merge. • BR-02: Survivorship defaults shall favor highest trust, latest reviewed, and most complete values, in that order unless admin overrides apply. • BR-03: Unmerge is an exceptional administrative action and must preserve full history.
Edge Cases	• One subject may legitimately share address, bank account, or family phone with others; these alone cannot force merge. • Previously merged records may later require split due to misidentification.
Failure Handling	• If downstream re-linking fails during merge, transaction shall roll back or complete through resumable compensating job with audit trail. • If search index update fails post-merge, canonical DB state shall remain source of truth and index rebuild queued.

FR-26 - Model governance, training, validation, and deployment management

Field	Detail
Priority	Must Have
Target Release	R1
RFP Traceability	RFP-P2-07, RFP-TECH-02, RFP-TECH-03
Description	The platform shall manage AI model versions, training datasets, prompts, validation results, production approvals, and rollback decisions for all ML and LLM-assisted features.
User Stories	• As a model governance reviewer, I want model versions and validation evidence tracked so that only approved models run in production.
Acceptance Criteria (Testable)	• AC-01: System shall register every model, prompt template, extraction schema, and ruleset version used in production or testing. • AC-02: Training and benchmark metadata shall record dataset identifiers, sample counts, evaluation metrics, run date, reviewer, and approval status. • AC-03: Only APPROVED model or prompt versions may be used by production workflows. • AC-04: The platform shall support rollback to a prior approved version without redeploying the entire application. • AC-05: If benchmark thresholds are not met, the associated feature shall switch to manual-review or rule-only mode according to configuration.
Business Rules	• BR-01: Production promotion requires named approver and stored evaluation artifact. • BR-02: Prompts for governed outputs must use schema-constrained responses and evidence citation instructions. • BR-03: Model artifacts and datasets shall remain on Department-controlled infrastructure only.
Edge Cases	• One feature may use multiple sub-models such as OCR, classifier, reranker, and summarizer; each must be versioned independently. • Prompt-only changes without model changes are still governance-controlled versions.
Failure Handling	• If a newly promoted model produces operational failures above threshold, automatic rollback or manual rollback procedure shall be available. • If validation artifact storage fails, approval cannot be completed.

5. Data Model Requirements

5.1 Canonical 54-Column Subject Profile Schema

Implementation note: the 54-column view is the mandated business output. The physical database should normalize repeated structures such as gang associates, documents, phones, bank accounts, and social media references into child tables or JSON sub-schemas, then expose the 54-column view through an API projection or database view for reporting and interoperability.

Entity	Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
ENT-SUBJ	district	Operational or case jurisdiction district	varchar(100)	Y	Must map to active district master value	CCTNS/FIR/MR/ Lead	RFP-P1-01
ENT-SUBJ	police_station	Police station linked to case or subject context	varchar(150)	N	Must map to police station master when present	CCTNS/FIR	RFP-P1-01
ENT-SUBJ	crime_number	Crime number / case reference	varchar(50)	N	Pattern configurable by district/year; unique within police_station+year when case-linked	CCTNS/FIR	RFP-P1-01
ENT-SUBJ	section_of_law	Applicable legal sections	text[]	N	Each value must exist in active legal section master	FIR/Legal review	RFP-BIZ-05
ENT-SUBJ	name	Subject full name	varchar(200)	N	Unicode; trim extra whitespace	FIR/CCTNS/IR/ Lead	RFP-P1-01
ENT-SUBJ	aliases	Known aliases and alternate spellings	text[]	N	Max 50 aliases; duplicate values not allowed	CCTNS/FIR/IR/ OSINT	RFP-P2-01
ENT-SUBJ	father_name	Father or guardian name	varchar(200)	N	Unicode	FIR/CCTNS/IR	RFP-P1-01
ENT-SUBJ	dob	Date of birth	date	N	Cannot be in future	CCTNS/ID docs	RFP-P1-01
ENT-SUBJ	age	Age when DOB unavailable	integer	N	0-120	FIR/CCTNS/IR	RFP-P1-01
ENT-SUBJ	nationality	Nationality	varchar(100)	N	Must map to nationality master where configured	Passport/CCTNS/ FIR	RFP-P1-01
ENT-SUBJ	occupation	Occupation or profession	varchar(150)	N	Free text limited to 150 chars or occupation master	FIR/IR	RFP-P1-01
ENT-SUBJ	residential_address	Current residential address	text	N	Store raw plus normalized form	FIR/CCTNS/Lead	RFP-P1-01
ENT-SUBJ	native_or_permanent_address	Native or permanent address	text	N	Store raw plus normalized form	FIR/IR	RFP-P1-01
ENT-SUBJ	native_state	Native state	varchar(100)	N	State master when present	FIR/IR	RFP-P1-01
ENT-SUBJ	mobile_numbers	Linked mobile numbers	text[]	N	Normalize to E.164 or national canonical format; duplicates removed	CDR/CDAT/FIR/ Lead	RFP-P1-01
ENT-SUBJ	aadhaar_number	Aadhaar number	varchar(12)	N	12 digits; encrypted at rest and masked in UI	ID docs/CCTNS	RFP-P1-01
ENT-SUBJ	pan_number	PAN number	varchar(10)	N	Regex [A-Z]{5}[0-9]{4}[A-Z]; encrypted and masked	ID docs/Financial docs	RFP-P1-01
ENT-SUBJ	ration_card_number	Ration card reference	varchar(50)	N	Alphanumeric pattern configurable	ID docs	RFP-P1-01
ENT-SUBJ	vehicle_rc_details	Vehicle RC details	jsonb	N	Validated against vehicle sub-schema	Transport docs/FIR	RFP-P1-01
ENT-SUBJ	driving_license_details	Driving license details	jsonb	N	Validated against DL sub-schema	ID docs/Transport	RFP-P1-01
ENT-SUBJ	identification_marks	Physical identification marks	text	N	Free text up to 2000 chars	CCTNS/IR	RFP-P1-01
ENT-SUBJ	photo_references	Linked photo evidence references	text[]	N	Must reference existing evidence IDs or URIs	CCTNS/evidence	RFP-P1-01

Entity	Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
ENT-SUBJ	bank_account_details	Linked bank accounts	jsonb	N	Validated against account sub-schema; encrypted fields	Financial docs/Unocross	RFP-P1-01
ENT-SUBJ	transaction_mode	Observed transaction mode	enum(transaction_mode)	N	Allowed values CASH, UPI, BANK_TRANSFER, MIXED, OTHER	Financial docs	RFP-P1-09
ENT-SUBJ	bank_statement_available	Bank statement availability flag	boolean	N	Default false	Financial docs/analyst review	RFP-P1-10
ENT-SUBJ	cdr_status	CDR request/processing status	enum(cdr_status)	Y	Allowed values NOT_REQUESTED, REQUESTED, RECEIVED, PROCESSED, UNAVAILABLE	CDR pipeline	RFP-P1-03
ENT-SUBJ	cdat_links	Links or references to CDAT data	text[]	N	Must reference CDAT artifacts or URLs	CDAT	RFP-P1-03
ENT-SUBJ	dopams_links	Internal related DOPAMS links	text[]	N	Must reference existing internal record IDs	DOPAMS internal	RFP-P1-01
ENT-SUBJ	offender_status	Current offender status	enum(offender_status)	Y	Allowed values ARRESTED, ABSCONDING, DETAINED, UNDER_INQUIRY, UNKNOWN	CCTNS/FIR/legal review	RFP-P1-01
ENT-SUBJ	offender_role	Active and historic offender role(s)	enum(offender_role)[]	N	Allowed values from role master	Classifier/manual review	RFP-P2-06
ENT-SUBJ	drug_procurement_method	Drug procurement method narrative	text	N	Source-cited narrative; max 4000 chars	FIR/Confession/NLP	RFP-P1-01
ENT-SUBJ	drug_delivery_method	Drug delivery method narrative	text	N	Source-cited narrative; max 4000 chars	FIR/Confession/NLP	RFP-P1-01
ENT-SUBJ	pd_act_details	PD Act details	text	N	Free text or structured child table reference	Legal docs	RFP-P1-01
ENT-SUBJ	history_sheet_details	History sheet details	text	N	Free text or structured child table reference	Police records	RFP-P1-01
ENT-SUBJ	fit_for_68f	Eligibility flag for Section 68F-related action	boolean	N	Default false; reviewer controlled	Legal review	RFP-P1-01
ENT-SUBJ	fit_for_pitndps_act	Eligibility flag for PITNDPS-related action	boolean	N	Default false; reviewer controlled	Legal review	RFP-P1-01
ENT-SUBJ	social_media_links	Approved social media URLs/handles	text[]	N	Valid URL/handle format; platform tagged in child table for physical schema	OSINT/monitored content	RFP-BIZ-02
ENT-SUBJ	passport_details	Passport metadata	jsonb	N	Validated against passport sub-schema; encrypted fields	Passport/IVFRT	RFP-P1-04
ENT-SUBJ	visa_details	Visa metadata	jsonb	N	Validated against visa sub-schema; encrypted fields	Passport/IVFRT	RFP-P1-04
ENT-SUBJ	gang_associate_1	Gang associate reference slot 1	varchar(200)	N	Canonical view only; physical schema normalized to associate child table	Investigation data	RFP-P1-01
ENT-SUBJ	gang_associate_2	Gang associate reference slot 2	varchar(200)	N	Canonical view only; physical schema normalized to associate child table	Investigation data	RFP-P1-01
ENT-SUBJ	gang_associate_3	Gang associate reference slot 3	varchar(200)	N	Canonical view only; physical schema normalized to	Investigation data	RFP-P1-01

Entity	Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
					associate child table		
ENT-SUBJ	gang_associate_4	Gang associate reference slot 4	varchar(200)	N	Canonical view only; physical schema normalized to associate child table	Investigation data	RFP-P1-01
ENT-SUBJ	gang_associate_5	Gang associate reference slot 5	varchar(200)	N	Canonical view only; physical schema normalized to associate child table	Investigation data	RFP-P1-01
ENT-SUBJ	gang_associate_6	Gang associate reference slot 6	varchar(200)	N	Canonical view only; physical schema normalized to associate child table	Investigation data	RFP-P1-01
ENT-SUBJ	gang_associate_7	Gang associate reference slot 7	varchar(200)	N	Canonical view only; physical schema normalized to associate child table	Investigation data	RFP-P1-01
ENT-SUBJ	gang_associate_8	Gang associate reference slot 8	varchar(200)	N	Canonical view only; physical schema normalized to associate child table	Investigation data	RFP-P1-01
ENT-SUBJ	gang_associate_9	Gang associate reference slot 9	varchar(200)	N	Canonical view only; physical schema normalized to associate child table	Investigation data	RFP-P1-01
ENT-SUBJ	gang_associate_10	Gang associate reference slot 10	varchar(200)	N	Canonical view only; physical schema normalized to associate child table	Investigation data	RFP-P1-01
ENT-SUBJ	gang_associate_11	Gang associate reference slot 11	varchar(200)	N	Canonical view only; physical schema normalized to associate child table	Investigation data	RFP-P1-01
ENT-SUBJ	whatsapp_chat_references	WhatsApp chat evidence references	text[]	N	Must reference preserved evidence items	Evidence/ approved exports	RFP-P1-01
ENT-SUBJ	social_media_chat_references	Social media chat evidence references	text[]	N	Must reference preserved evidence items	Evidence/ approved exports	RFP-P1-01
ENT-SUBJ	source_document_references	Source document/evidence IDs supporting profile	text[]	Y	At least one evidence reference required for approved profile publication	All approved sources	RFP-BIZ-06
ENT-SUBJ	extraction_confidence_score	Aggregate population confidence score	numeric(5,4)	Y	0.0000-1.0000	System derived	RFP-P1-02

5.2 Supporting Entity Schemas

ENT-CASE

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
case_id	Canonical case UUID	uuid	Y	Primary key	System	RFP-P1-01
crime_number	Crime number reference	varchar(50)	Y	Unique within police_station+year	CCTNS/FIR	RFP-P1-01
district	District	varchar(100)	Y	District master value	CCTNS/FIR	RFP-P1-01
police_station	Police station	varchar(150)	Y	Police station master	CCTNS/FIR	RFP-P1-01

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
				value		
sections_of_law	Current sections applied	text[]	N	Legal section master values only	FIR/Legal review	RFP-BIZ-05
case_status	Case lifecycle status	enum(case_status)	Y	OPEN, UNDER_INVESTIGATION, CHARGESHEETED, CLOSED, ARCHIVED	Case workflow	RFP-P1-08
primary_subject_id	Primary linked subject	uuid	N	Foreign key to subject	System	RFP-P2-08
opened_at	Case opened timestamp	timestamptz	N	Cannot be future	CCTNS/FIR	RFP-P1-01

ENT-DOC

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
document_id	Evidence/document UUID	uuid	Y	Primary key	System	RFP-BIZ-06
source_system	Originating source	enum(source_system)	Y	Must be active source master value	Connector/manual upload	RFP-P1-03
document_type	Document type	enum(document_type)	Y	Must be active document master value	Connector/manual upload	RFP-P1-03
storage_uri	Immutable object storage URI	text	Y	Must resolve to evidence storage path	System	RFP-BIZ-06
checksum_sha256	SHA-256 hash of stored artifact	char(64)	Y	64 lowercase hex chars	System	RFP-BIZ-06
language	Detected or declared language	enum(language_code)	N	EN, TE, MIXED, UNKNOWN	System/manual	RFP-P1-13
ingest_job_id	Originating ingest job	uuid	Y	Foreign key to ingest job	System	RFP-P1-03
legal_hold_flag	Legal hold status	boolean	Y	Default false	Legal reviewer	RFP-BIZ-06

ENT-EXT

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
extraction_id	Extraction row UUID	uuid	Y	Primary key	System	RFP-P1-13
document_id	Parent document	uuid	Y	Foreign key to document	System	RFP-P1-13
field_name	Extracted field name	varchar(100)	Y	Must exist in extraction schema	System	RFP-P1-13
machine_value	Machine extracted value	text	N	Schema-validated	System	RFP-P1-13
reviewer_value	Reviewer-confirmed value	text	N	Schema-validated	Reviewer	RFP-P1-13
confidence_score	Model confidence	numeric(5,4)	N	0.0000-1.0000	System	RFP-P1-13
review_status	Review disposition	enum(review_status)	Y	AUTO_ACCEPTED, REVIEW_REQUIRED, ACCEPTED, CORRECTED, REJECTED	Workflow	RFP-P1-13
source_ref	Page/span/bbox reference	jsonb	N	Validated against source_ref schema	System	RFP-P1-13

ENT-LEGAL

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
order_id	Legal order UUID	uuid	Y	Primary key	System	RFP-P1-08
case_id	Linked case	uuid	Y	Foreign key to case	System	RFP-P1-08
court_name	Court name	varchar(255)	Y	Non-empty	E-Courts	RFP-P1-08
order_type	Order type	enum(order_type)	Y	JUDGMENT, BAIL_ORDER, INTERIM_ORDER, OTHER	E-Courts	RFP-P1-08
order_date	Order date	date	Y	Cannot be future	E-Courts	RFP-P1-08
status_effect	Resulting legal status effect	varchar(100)	N	Mapped via legal status master	System/reviewer	RFP-P1-08
retrieved_at	Retrieved timestamp	timestamptz	Y	System timestamp	System	RFP-P1-08
match_confidence	Match confidence	numeric(5,4)	N	0.0000-1.0000	System	RFP-P1-08

ENT-COMM

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
event_id	Communication event UUID	uuid	Y	Primary key	System	RFP-P1-03
source_system	Telecom source	enum(source_system)	Y	CDR, CDAT, IPDR or approved source	Connector	RFP-P1-03
from_number	Calling/source number	varchar(20)	Y	Normalized telecom format	CDR/CDAT/IPDR	RFP-P2-02
to_number	Receiving/target number	varchar(20)	Y	Normalized telecom format	CDR/CDAT/IPDR	RFP-P2-02
start_time	Event start	timestamptz	Y	Valid timestamp	CDR/CDAT/IPDR	RFP-P2-02
duration_seconds	Duration in seconds	integer	N	0 or greater	CDR/CDAT	RFP-P2-03
cell_tower_ref	Tower/location reference	varchar(100)	N	Normalized tower master reference	CDAT/LBS	RFP-P2-03
linked_subject_ids	Resolved subject links	uuid[]	N	FK array or relationship table	System	RFP-P2-08

ENT-TXN

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
txn_id	Transaction UUID	uuid	Y	Primary key	System	RFP-P1-09
account_ref	Source account reference	varchar(100)	N	Encrypted/tokenized if sensitive	Financial source	RFP-P1-09
counterparty_ref	Counterparty account/UPI reference	varchar(150)	N	Encrypted/tokenized if sensitive	Financial source	RFP-P1-09
linked_phone	Linked phone number	varchar(20)	N	Normalized telecom format	Financial source/system	RFP-P1-09
txn_time	Transaction timestamp	timestamptz	Y	Valid timestamp	Financial source	RFP-P1-09
amount	Transaction amount	numeric(14,2)	N	>=0	Financial source	RFP-P1-09
txn_mode	Transaction mode	enum(transaction_mode)	Y	CASH, UPI, BANK_TRANSFER, MIXED, OTHER	Financial source	RFP-P1-09
match_status	Telecom/subject match status	enum(match_status)	Y	UNMATCHED, PARTIAL, MATCHED, REVIEW_REQUIRED	System	RFP-P1-09

ENT-DEV

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
device_event_id	Device history UUID	uuid	Y	Primary key	System	RFP-P2-03
number	Mobile number	varchar(20)	Y	Normalized telecom format	CDR/CDAT	RFP-P2-03
imei	IMEI	varchar(20)	N	14-16 digits/pattern	CDR/CDAT	RFP-P2-03
first_seen_at	First seen timestamp	timestamptz	N	Valid timestamp	CDR/CDAT	RFP-P2-03
last_seen_at	Last seen timestamp	timestamptz	N	>= first_seen_at	CDR/CDAT	RFP-P2-03
subject_id	Linked subject	uuid	N	Foreign key	System	RFP-P2-08

ENT-LOC

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
location_event_id	Location event UUID	uuid	Y	Primary key	System	RFP-P2-03
subject_id	Linked subject	uuid	N	Foreign key	System	RFP-P2-03
number	Linked mobile number	varchar(20)	N	Normalized telecom format	LBS/CDAT	RFP-P2-03
latitude	Latitude	numeric(9,6)	N	-90 to 90	LBS/GIS	RFP-P2-03
longitude	Longitude	numeric(9,6)	N	-180 to 180	LBS/GIS	RFP-P2-03
event_time	Event timestamp	timestamptz	Y	Valid timestamp	LBS/GIS	RFP-P2-03
accuracy_meters	Location accuracy	numeric(8,2)	N	>=0	LBS/GIS	RFP-P2-03
geofence_hits	Matched geofences	uuid[]	N	FK array or relationship table	System	RFP-P2-04

ENT-NODE

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
node_id	Network node UUID	uuid	Y	Primary key	System	RFP-P2-02
node_type	Node type	enum(node_type)	Y	SUBJECT, PHONE, ACCOUNT, DEVICE, VEHICLE, LOCATION,	System	RFP-P2-02

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
				CASE		
display_value	Display label	varchar(255)	Y	Non-empty	System/source	RFP-P2-02
canonical_ref	Reference to canonical entity	varchar(100)	N	Foreign key or external ref	System	RFP-P2-02
risk_score	Current node risk score	numeric(5,2)	N	0-100	System	RFP-BIZ-04

ENT-EDGE

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
edge_id	Network edge UUID	uuid	Y	Primary key	System	RFP-P2-02
node_a_id	Source node	uuid	Y	FK to network node	System	RFP-P2-02
node_b_id	Target node	uuid	Y	FK to network node	System	RFP-P2-02
edge_type	Relationship type	enum(edge_type)	Y	CALL, MESSAGE, TRANSACTION, CO_CASE, CO_LOCATION, DOCUMENT, OTHER	System	RFP-P2-02
weight	Computed relationship weight	numeric(10,4)	N	>=0	System	RFP-P2-02
evidence_count	Evidence count supporting edge	integer	N	>=0	System	RFP-P2-02
time_window	Aggregated time range	tstzrange	N	Valid range	System	RFP-P2-02

ENT-LEAD

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
lead_id	Lead UUID	uuid	Y	Primary key	System	RFP-P2-09
channel	Lead source channel	enum(lead_channel)	Y	TOLL_FREE, WHATSAPP, WALK_IN, FIELD, OTHER	Operator	RFP-P2-09
informant_name	Informant name	varchar(200)	N	May be null for anonymous leads	Operator	RFP-P2-09
informant_contact	Informant contact	varchar(50)	N	Normalized phone/email if present	Operator	RFP-P2-09
location_text	Reported location	text	Y	Non-empty	Operator	RFP-P2-09
suspect_info	Reported suspect details	text	N	Up to 4000 chars	Operator	RFP-P2-09
urgency	Urgency level	enum(urgency_level)	Y	LOW, MEDIUM, HIGH, CRITICAL	Operator	RFP-P2-09
status	Lead status	enum(lead_status)	Y	NEW, VALIDATED, MEMO_GENERATED, APPROVAL_PENDING, ROUTED, IN_ACTION, CLOSED, REJECTED	Workflow	RFP-P2-10

ENT-MEMO

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
memo_id	Memo UUID	uuid	Y	Primary key	System	RFP-P2-10
lead_id	Linked lead	uuid	N	FK to lead	System	RFP-P2-10
template_version_id	Applied template version	uuid	Y	FK to template version	System	RFP-P2-10
generated_at	Generated timestamp	timestamptz	Y	System timestamp	System	RFP-P2-10
approver_user_id	Approving user	uuid	N	FK to user	Workflow	RFP-P2-10
dispatch_status	Dispatch state	enum(dispatch_status)	Y	DRAFT, APPROVED, DISPATCHED, FAILED, CANCELLED	Workflow	RFP-P2-10
dispatch_channel	Dispatch channel	enum(dispatch_channel)	N	EMAIL, PRINT, DOWNLOAD, INTERNAL_QUEUE	Workflow	RFP-P2-10
output_document_id	Generated output document	uuid	N	FK to document	System	RFP-P2-10

ENT-ALERT

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
alert_id	Alert UUID	uuid	Y	Primary key	System	RFP-BIZ-07
alert_type	Alert type	enum(alert_type)	Y	GEOFENCE, RISK_SCORE, OVERDUE_REVIEW, CONNECTOR_FAILURE, CRITICAL_LEAD, OTHER	System	RFP-BIZ-07
severity	Severity	enum(severity_level)	Y	LOW, MEDIUM, HIGH, CRITICAL	System	RFP-BIZ-07
entity_ref	Linked entity reference	varchar(100)	Y	FK or typed reference	System	RFP-BIZ-07
triggered_at	Trigger timestamp	timestamptz	Y	Valid timestamp	System	RFP-BIZ-07
status	Alert status	enum(alert_status)	Y	OPEN, ACKNOWLEDGED, ASSIGNED, ESCALATED, RESOLVED, CLOSED, FALSE_POSITIVE	Workflow	RFP-BIZ-07
assigned_to	Current assignee	uuid	N	FK to user	Workflow	RFP-BIZ-07
sla_due_at	SLA due timestamp	timestamptz	N	Valid timestamp	Workflow	RFP-BIZ-07

ENT-WFTASK

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
task_id	Workflow task UUID	uuid	Y	Primary key	System	RFP-BIZ-07
task_type	Workflow task type	enum(task_type)	Y	REVIEW, APPROVAL, ESCALATION, REPROCESS, CONFIG_APPROVAL, MODEL_APPROVAL	Workflow	RFP-BIZ-07
linked_entity_type	Linked entity type	varchar(50)	Y	Controlled entity list	System	RFP-BIZ-07
linked_entity_id	Linked entity ID	uuid	Y	Foreign key by entity type	System	RFP-BIZ-07
status	Task status	enum(task_status)	Y	OPEN, IN_PROGRESS, WAITING, COMPLETED, CANCELLED, EXPIRED	Workflow	RFP-BIZ-07
owner_user_id	Task owner	uuid	N	FK to user	Workflow	RFP-BIZ-07
due_at	Due timestamp	timestamptz	N	Valid timestamp	Workflow	RFP-BIZ-07

ENT-USER

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
user_id	User UUID	uuid	Y	Primary key	System	RFP-TECH-07
username	Login identifier	varchar(150)	Y	Unique	IdP/System	RFP-TECH-07
role_id	Assigned role	varchar(20)	Y	Must exist in role master	Admin	RFP-TECH-07
jurisdiction_scope	Jurisdiction scope JSON	jsonb	Y	Validated against scope schema	Admin	RFP-TECH-07
auth_provider	Identity provider	enum(auth_provider)	Y	SSO, LDAP, AD, LOCAL_BREAK_GLASS	System	RFP-TECH-07
active_status	User active flag	boolean	Y	Default true	Admin	RFP-TECH-07

ENT-AUDIT

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
audit_id	Audit row UUID	uuid	Y	Primary key	System	RFP-BIZ-06
actor_user_id	Acting user	uuid	N	FK to user or null for service account event	System	RFP-TECH-07
action_type	Action type	varchar(100)	Y	Controlled action catalog	System	RFP-BIZ-06
target_type	Target entity type	varchar(50)	Y	Controlled entity list	System	RFP-BIZ-06
target_id	Target entity ID	varchar(100)	Y	FK or external reference	System	RFP-BIZ-06
event_time	Event timestamp	timestamptz	Y	System timestamp	System	RFP-BIZ-06
before_after_snapshot	Changed fields snapshot	jsonb	N	JSON diff schema	System	RFP-BIZ-06
correlation_id	Correlation ID	varchar(100)	N	Traceable request/job	System	RFP-BIZ-06

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
				correlation		

ENT-MODEL

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
model_version_id	Model version UUID	uuid	Y	Primary key	System	RFP-P2-07
model_name	Model or prompt name	varchar(150)	Y	Non-empty	System	RFP-TECH-03
task_type	Feature/task type	enum(ai_task_type)	Y	OCR, EXTRACTION, CLASSIFICATION, SUMMARY, RISK_SCORING, LEGAL_MAPPING, QA	System	RFP-TECH-03
version_label	Semantic version	varchar(50)	Y	Unique per model_name	System	RFP-P2-07
validation_score	Primary validation metric	numeric(6,4)	N	0.0000-1.0000	Evaluation pipeline	RFP-P2-07
approval_status	Promotion status	enum(approval_status)	Y	DRAFT, TESTED, APPROVED, REJECTED, RETIRED	Workflow	RFP-P2-07
artifact_uri	Stored artifact location	text	Y	Department-controlled storage only	System	RFP-TECH-02
approved_at	Approval timestamp	timestamptz	N	System timestamp	Workflow	RFP-P2-07

ENT-CONN

Attribute	Description	Data Type	Mandatory (Y/N)	Validation Rule	Source	RFP Ref
connector_id	Connector UUID	uuid	Y	Primary key	System	RFP-P1-04
connector_name	Connector name	varchar(150)	Y	Unique	Admin	RFP-P1-04
source_system	Mapped source system	enum(source_system)	Y	Must exist in source master	Admin	RFP-P1-04
interface_type	Interface type	enum(interface_type)	Y	API, SFTP, DB_LINK, FILE_WATCH, WEBHOOK, SCRAPE	Admin	RFP-P1-04
auth_type	Authentication type	enum(auth_type)	Y	OAUTH, BASIC, API_KEY, MTLS, DB_CREDENTIAL, NONE	Admin	RFP-P1-04
feature_flag_enabled	Connector active flag	boolean	Y	Default false for optional connectors	Admin	RFP-P1-04
sync_schedule	Cron or interval	varchar(100)	N	Valid schedule format	Admin	RFP-P1-04
last_health_status	Connector health	enum(health_status)	N	HEALTHY, DEGRADED, FAILED, DISABLED	System	RFP-P1-04

5.3 Master Data and Enumerations

Enumeration ID	Enumeration	Allowed Values	Used By
ENUM-001	transaction_mode	CASH | UPI | BANK_TRANSFER | MIXED | OTHER	SubjectProfile, FinancialTransaction
ENUM-002	cdr_status	NOT_REQUESTED | REQUESTED | RECEIVED | PROCESSED | UNAVAILABLE	SubjectProfile
ENUM-003	offender_status	ARRESTED | ABSCONDING | DETAINED | UNDER_INQUIRY | UNKNOWN	SubjectProfile
ENUM-004	offender_role	CULTIVATOR | MANUFACTURER | SUPPLIER | PEDDLER | TRANSPORTER | CONSUMER | MEDIATOR | FINANCIER	SubjectProfile, classifier
ENUM-005	case_status	OPEN | UNDER_INVESTIGATION | CHARGESHEETED | CLOSED | ARCHIVED	Case
ENUM-006	review_status	AUTO_ACCEPTED | REVIEW_REQUIRED | ACCEPTED | CORRECTED | REJECTED	ExtractionField
ENUM-007	lead_status	NEW | VALIDATED | MEMO_GENERATED | APPROVAL_PENDING | ROUTED | IN_ACTION | CLOSED | REJECTED	Lead
ENUM-008	alert_status	OPEN | ACKNOWLEDGED | ASSIGNED | ESCALATED | RESOLVED | CLOSED | FALSE_POSITIVE	Alert
ENUM-009	severity_level	LOW | MEDIUM | HIGH | CRITICAL	Alert, task
ENUM-010	task_status	OPEN | IN_PROGRESS | WAITING | COMPLETED | CANCELLED | EXPIRED	WorkflowTask

Enumeration ID	Enumeration	Allowed Values	Used By
ENUM-011	task_type	REVIEW | APPROVAL | ESCALATION | REPROCESS | CONFIG_APPROVAL | MODEL_APPROVAL	WorkflowTask
ENUM-012	dispatch_status	DRAFT | APPROVED | DISPATCHED | FAILED | CANCELLED	Memo/Requisition
ENUM-013	source_system	CCTNS | CDAT | CDR | IPDR | FIR_REPOSITORY | CONFESSION_MEMO | IR_REPOSITORY | E_SAKSHYA | C_TRACE | GOOGLE_DRIVE | E_COURTS | TS_COP | ICJS | TRANSPORT | E_PRISONS | IVFRT | NHAI | LAND_STAMP | MANUAL_UPLOAD | OTHER	Multiple
ENUM-014	document_type	FIR | PANCHANAMA | CONFESSION_MEMO | INTERROGATION_REPORT | MONTHLY_REPORT | COURT_ORDER | SCREENSHOT | CHAT_EXPORT | BANK_STATEMENT | CDR_FILE | IPDR_FILE | TECHNICAL_REPORT | DOSSIER | MEMO | OTHER	SourceDocument
ENUM-015	lead_channel	TOLL_FREE | WHATSAPP | WALK_IN | FIELD | EMAIL | OTHER	Lead
ENUM-016	approval_status	DRAFT | TESTED | APPROVED | REJECTED | RETIRED	ModelVersion/configs
ENUM-017	auth_provider	SSO | LDAP | AD | LOCAL_BREAK_GLASS	User
ENUM-018	interface_type	API | SFTP | DB_LINK | FILE_WATCH | WEBHOOK | SCRAPE	ConnectorConfig
ENUM-019	health_status	HEALTHY | DEGRADED | FAILED | DISABLED	ConnectorConfig
ENUM-020	urgency_level	LOW | MEDIUM | HIGH | CRITICAL	Lead

5.4 Physical Modeling Notes

- Use UUID primary keys for all canonical entities.
- Store all timestamps in timestamp with time zone and display in Asia/Kolkata by default.
- Use PostgreSQL jsonb only for bounded sub-schemas that are validated in application logic; do not use free-form JSON as a substitute for core relational design.
- Preserve raw and derived evidence as separate artefacts with parent-child lineage.
- Use append-only assertion/version tables for extracted or reviewer-corrected values rather than destructive overwrites.

6. API Specifications

All APIs shall be versioned under /api/v1, enforce server-side authorization, emit correlation IDs, and return standard error payloads in the form {code, message, correlation_id, details[]}.

API-01 - Create manual ingestion job

Field	Detail
Method	POST
Endpoint	/api/v1/ingestion-jobs
Authentication	Bearer token via Department SSO; roles PL-1 and above.
Rate Limits	60 requests/minute/user; 500 requests/minute/system.
Idempotency Handling	Required. Idempotency-Key header prevents duplicate job creation for the same upload.
Validation Rules	source_system, document_type, and object_key are mandatory. object_key must point to an existing staged upload. case_ref fields must match configured patterns when provided.
Error Codes	400 VALIDATION_ERROR, 401 UNAUTHENTICATED, 403 FORBIDDEN, 409 DUPLICATE_PAYLOAD, 422 UNSUPPORTED_DOCUMENT_TYPE, 500 INTERNAL_ERROR

Request Schema (Example):

```
{
  "source_system": "MANUAL_UPLOAD",
  "document_type": "FIR",
  "object_key": "evidence/raw/2026/03/02/fir-001.pdf",
  "district": "Hyderabad",
  "police_station": "Banjara Hills",
  "case_ref": {
    "crime_number": "123/2026"
  },
  "language_hint": "TE",
  "tags": ["phase1", "fir"]
}
```

Response Schema (Example):

```
{
  "job_id": "7c6b3d66-3f49-4d10-9fa3-2df9fd01b90a",
  "status": "RECEIVED",
  "queued_at": "2026-03-02T10:20:00+05:30",
  "correlation_id": "ing-20260302-000123"
}
```

API-02 - Get ingestion job status

Field	Detail
Method	GET
Endpoint	/api/v1/ingestion-jobs/{job_id}
Authentication	Bearer token; roles PL-1 and above.
Rate Limits	300 requests/minute/user.
Idempotency Handling	Not applicable.
Validation Rules	job_id must be a valid UUID visible within the requestor's jurisdiction.
Error Codes	401, 403, 404 JOB_NOT_FOUND, 500

Request Schema (Example):

```
{"path_params":{"job_id":"uuid"}}
```

Response Schema (Example):

```
{
  "job_id": "7c6b3d66-3f49-4d10-9fa3-2df9fd01b90a",
  "status": "REVIEW_REQUIRED",
  "source_system": "MANUAL_UPLOAD",
  "document_id": "2df2f4f1-76ff-4326-9e7d-11582c76f2b9",
  "issues": ["LOW_OCR_CONFIDENCE"],
  "retry_count": 1
}
```

API-03 - Submit extraction review decision

Field	Detail
Method	POST
Endpoint	/api/v1/extractions/{extraction_id}/review
Authentication	Bearer token; PL-1 and above within jurisdiction.
Rate Limits	120 requests/minute/user.
Idempotency Handling	Optional; duplicate submissions with same payload within 30 seconds return same result.
Validation Rules	decision must be ACCEPTED, CORRECTED, or REJECTED. reviewer_value is mandatory for CORRECTED. reason_code mandatory when decision differs from machine value.
Error Codes	400, 401, 403, 404, 409 INVALID_STATE, 422 SCHEMA_VALIDATION_FAILED

Request Schema (Example):

```
{
  "decision": "CORRECTED",
  "reviewer_value": "Section 8(c) NDPS Act",
  "reason_code": "OCR_MISREAD",
```

```
  "comment": "Verified against page 2"
}
```


Response Schema (Example):


```
{
  "extraction_id": "0c77e337-0fa1-4ab7-b7ca-1b8c6e7d4b0d",
  "review_status": "CORRECTED",
  "reviewed_at": "2026-03-02T10:42:11+05:30",
  "next_state": "PROFILE_UPDATE_PENDING"
}
```


API-04 - Search subjects and entities

Field	Detail
Method	POST
Endpoint	/api/v1/subjects/search
Authentication	Bearer token; PL-2 and above for advanced search; restricted identifier searches require role-specific permission.
Rate Limits	90 requests/minute/user.
Idempotency Handling	Not applicable.
Validation Rules	query is mandatory. query_type must be in approved search modes. page_size max 100. Sensitive query types require elevated roles.
Error Codes	400, 401, 403, 422 INVALID_QUERY_TYPE, 429 RATE_LIMITED, 500

Request Schema (Example):


```
{
  "query": "Raju",
  "query_type": "NAME",
  "filters": {
    "district": ["Hyderabad"],
    "date_from": "2025-01-01",
    "date_to": "2026-03-02"
  },
  "include_masked_hits": false,
  "page": 1,
  "page_size": 25
}
```


Response Schema (Example):

```
{
  "results": [
    {
      "subject_id": "9f8d11d3-2e2f-4c6e-b9fa-4ba7a9e1d021",
      "display_name": "Raju @ Rajesh",
      "match_score": 0.97,
      "match_reasons": ["exact_alias", "same_mobile"],
      "district": "Hyderabad",
      "masked_identifiers": {
        "aadhaar": "XXXX-XXXX-1234"
      }
    }
  ],
  "page": 1,
  "page_size": 25,
  "total": 1
}
```


API-05 - Get subject dossier

Field	Detail
Method	GET
Endpoint	/api/v1/subjects/{subject_id}
Authentication	Bearer token; PL-2 and above.
Rate Limits	120 requests/minute/user.
Idempotency Handling	Not applicable.
Validation Rules	subject_id must exist and be visible to the requestor. include_sections limited to approved values.
Error Codes	401, 403, 404 SUBJECT_NOT_FOUND, 500

Request Schema (Example):


```
{"path_params":{"subject_id":"uuid"},"query":{"include_sections":"cases,alerts,documents,network"}}
```


Response Schema (Example):


```
{
  "subject_id": "9f8d11d3-2e2f-4c6e-b9fa-4ba7a9e1d021",
  "profile_version": 14,
  "name": "Raju @ Rajesh",
  "offender_status": "ABSCONDING",
  "roles": ["PEDDLER"],
  "completeness_score": 0.82,
```

"cases": [{"case_id":"...","crime_number":"123/2026"}],
"recent_alerts": [{"alert_id":"...","severity":"HIGH"}]
}

API-06 - Generate interrogation report draft

Field	Detail
Method	POST
Endpoint	/api/v1/interrogation-reports/generate
Authentication	Bearer token; PL-1 create, PL-3 finalize.
Rate Limits	30 requests/minute/user.
Idempotency Handling	Recommended. Same case_id + template_version_id + source set within 5 minutes reuses existing draft unless force_regenerate=true.
Validation Rules	case_id and active template_version_id are mandatory. source_document_ids must belong to the case or authorized related entities.
Error Codes	400, 401, 403, 404 TEMPLATE_NOT_FOUND, 409 INVALID_TEMPLATE_STATE, 422 MANDATORY_SOURCE_MISSING, 500

Request Schema (Example):

{
"case_id": "4dc2f8f1-c74b-4ce8-a980-fda61910f1e3",
"template_version_id": "d18d3bb3-2150-4ba8-8c52-7e9f9cbec63f",
"source_document_ids": ["2df2f4f1-76ff-4326-9e7d-11582c76f2b9"],
"mode": "DRAFT"
}

Response Schema (Example):

{
"report_id": "f8046cc6-21cb-4f79-9d8b-5d6806c51a53",
"status": "REVIEW_REQUIRED",
"missing_fields": ["accused_address", "seizure_location"],
"document_preview_id": "91b8f463-6df5-44de-8f0e-c89e862f4d9c"
}

API-07 - Create network analysis job

Field	Detail
Method	POST
Endpoint	/api/v1/network-analysis/jobs
Authentication	Bearer token; PL-2 and above.
Rate Limits	10 job creations/minute/user.

Field	Detail
Idempotency Handling	Required for job creation to avoid duplicate heavy runs.
Validation Rules	At least one seed subject or seed number is required. depth max 5 or configured cap. date_from <= date_to.
Error Codes	400, 401, 403, 409 TOO_MANY_SEEDS, 422 DEPTH_LIMIT_EXCEEDED, 500

Request Schema (Example):

```
{
  "seed_subject_ids": ["9f8d11d3-2e2f-4c6e-b9fa-4ba7a9e1d021"],
  "depth": 3,
  "date_from": "2025-09-01",
  "date_to": "2026-03-02",
  "filters": {
    "min_duration_seconds": 30,
    "sources": ["CDR", "CDAT"]
  }
}
```

Response Schema (Example):

```
{
  "job_id": "ee2d20fa-0df8-4c2d-91cf-4d6d0baaf91f",
  "status": "QUEUED",
  "estimated_result_retention_days": 30
}
```

API-08 - Create lead

Field	Detail
Method	POST
Endpoint	/api/v1/leads
Authentication	Bearer token; PL-1 and above.
Rate Limits	120 requests/minute/user.
Idempotency Handling	Recommended for operator UI submissions.
Validation Rules	channel, location_text, and urgency are mandatory. informant_contact normalized when provided.
Error Codes	400, 401, 403, 422 INVALID_URGENCY, 500

Request Schema (Example):

```
{
  "channel": "WHATSAPP",
  "informant_name": "Confidential",
```

```
  "informant_contact": "+919999999999",
  "location_text": "Miyapur bus stop",
  "suspect_info": "Male, black bike, selling contraband near metro pillar",
  "urgency": "HIGH",
  "notes": "Received at 10:55 AM"
}
```


Response Schema (Example):


```
{
  "lead_id": "09df8b5c-45ea-4926-b4c1-c3fbbf9d53cd",
  "status": "NEW",
  "duplicate_candidates": [],
  "next_state": "VALIDATION_PENDING"
}
```


API-09 - Approve memo or requisition

Field	Detail
Method	POST
Endpoint	/api/v1/memos/{memo_id}/approve
Authentication	Bearer token; PL-3 and above.
Rate Limits	60 requests/minute/user.
Idempotency Handling	Required. Duplicate approval attempts return the current approved state.
Validation Rules	decision must be APPROVE or REJECT. dispatch_channel required for approval when auto-dispatch configured. Only current approver may act.
Error Codes	400, 401, 403, 404 MEMO_NOT_FOUND, 409 INVALID_STATE, 500

Request Schema (Example):


```
{
  "decision": "APPROVE",
  "dispatch_channel": "EMAIL",
  "comment": "Send to zonal officer immediately"
}
```


Response Schema (Example):


```
{
  "memo_id": "49dfb4f4-66ea-4b0a-8b1d-f4446c480100",
  "dispatch_status": "APPROVED",
  "approved_by": "user-123",
```

"approved_at": "2026-03-02T11:05:00+05:30"
}

API-10 - Acknowledge or assign alert

Field	Detail
Method	POST
Endpoint	/api/v1/alerts/{alert_id}/acknowledge
Authentication	Bearer token; PL-2 and above for acknowledge; PL-3 and above for escalate/close.
Rate Limits	120 requests/minute/user.
Idempotency Handling	Optional; repeated same action on same state returns current status.
Validation Rules	action must be ACKNOWLEDGE, ASSIGN, ESCALATE, RESOLVE, CLOSE, or FALSE_POSITIVE. assignee_user_id mandatory for ASSIGN.
Error Codes	400, 401, 403, 404 ALERT_NOT_FOUND, 409 INVALID_STATE, 500

Request Schema (Example):

{
"action": "ASSIGN",
"assignee_user_id": "7d43ef53-8e11-46ec-aa2d-c9859220f112",
"comment": "Field verification required"
}

Response Schema (Example):

{
"alert_id": "17a3bc0d-1e8e-49dc-a1e7-38f45f54f58d",
"status": "ASSIGNED",
"assigned_to": "7d43ef53-8e11-46ec-aa2d-c9859220f112",
"sla_due_at": "2026-03-02T11:20:00+05:30"
}

API-11 - Create or update governed rule/config version

Field	Detail
Method	POST
Endpoint	/api/v1/admin/config/rules
Authentication	Bearer token; PL-4 and above.
Rate Limits	30 requests/minute/user.
Idempotency Handling	Recommended for admin UI save actions.
Validation Rules	config_type, name, version_label, and payload are mandatory. payload validated against schema for config_type.
Error Codes	400, 401, 403, 409 VERSION_ALREADY_EXISTS, 422 SCHEMA_VALIDATION_FAILED, 500

Request Schema (Example):


```
{
  "config_type": "RISK_SCORING_RULESET",
  "name": "default-risk-v1",
  "version_label": "1.0.0",
  "effective_from": "2026-03-05T00:00:00+05:30",
  "payload": {
    "weights": {
      "source_credibility": 20,
      "linked_offender": 25,
      "keyword_pattern": 15,
      "repeat_occurrence": 20,
      "geofence_relevance": 20
    }
  }
}
```


Response Schema (Example):


```
{
  "config_version_id": "ab73a68e-6286-41c4-b989-bd8898e2e740",
  "status": "DRAFT",
  "approval_required": true
}
```


API-12 - Create report export job

Field	Detail
Method	POST
Endpoint	/api/v1/reports/export
Authentication	Bearer token; report-specific permission required.
Rate Limits	20 jobs/minute/user.
Idempotency Handling	Required for same report and identical filter payload within 2 minutes.
Validation Rules	report_id and format are mandatory. Only approved formats allowed for the selected report.
Error Codes	400, 401, 403, 404 REPORT_NOT_FOUND, 422 FORMAT_NOT_ALLOWED, 500

Request Schema (Example):


```
{
  "report_id": "REP-005",
  "format": "PDF",
  "filters": {
```

```
    "district": ["Hyderabad"],
    "date_from": "2026-02-01",
    "date_to": "2026-02-29"
  }
}
```


Response Schema (Example):


```
{
  "export_job_id": "56c81166-9f37-4251-90d2-2dc5e95d752f",
  "status": "QUEUED",
  "watermark": "Generated by user-123 on 2026-03-02 11:10 IST"
}
```


API-13 - Natural-language query

Field	Detail
Method	POST
Endpoint	/api/v1/ai/query
Authentication	Bearer token; PL-2 and above; data returned limited by RBAC.
Rate Limits	20 requests/minute/user.
Idempotency Handling	Not applicable.
Validation Rules	question mandatory; max length 2000 chars. response_mode optional but constrained to approved modes.
Error Codes	400, 401, 403, 422 UNSUPPORTED_QUERY, 429 RATE_LIMITED, 503 AI_SERVICE_UNAVAILABLE

Request Schema (Example):


```
{
  "question": "Show repeat offenders in Hyderabad with new bail orders in the last 30 days",
  "response_mode": "SUMMARY_WITH_TABLE"
}
```


Response Schema (Example):


```
{
  "answer_text": "Found 12 repeat offenders in Hyderabad with new bail orders in the last 30 days.",
  "structured_result": {
    "count": 12,
    "rows": [{"subject_id":"...","name":"...","order_date":"2026-02-20"}]
  },
  "citations": [
```

{"entity_type":"LegalOrder","entity_id":"..."},
{"entity_type":"Case","entity_id":"..."}
],
"model_version_id": "0f1d..."
}

API-14 - Verify evidence integrity

Field	Detail
Method	POST
Endpoint	/api/v1/evidence/{document_id}/hash-verify
Authentication	Bearer token; PL-3 and above, or auditor role.
Rate Limits	30 requests/minute/user.
Idempotency Handling	Not applicable.
Validation Rules	document_id must exist. Only evidence accessible to the role may be verified.
Error Codes	401, 403, 404 DOCUMENT_NOT_FOUND, 409 DOCUMENT_UNAVAILABLE, 500

Request Schema (Example):

{
"verify_mode": "RECALCULATE"
}

Response Schema (Example):

{
"document_id": "2df2f4f1-76ff-4326-9e7d-11582c76f2b9",
"stored_hash": "3df7...",
"calculated_hash": "3df7...",
"match": true,
"verified_at": "2026-03-02T11:15:00+05:30"
}

7. Workflow / State Machine

The workflow engine shall store every state transition with actor, timestamp, correlation ID, previous state, next state, and reason code. Time-based escalations shall be processed by a scheduler and shall be re-evaluated after priority changes.

Workflow ID	Workflow Name	Current State	Trigger	Condition	Next State	Actor	SLA Timer	Escalation	Notes
WF-ING-01	Ingestion and extraction	RECEIVED	Payload stored	Checksum valid	QUEUED	System	Immediate	None	Job created after upload/connector receipt.
WF-ING-01	Ingestion and	QUEUED	Worker starts	Resources available	PROCESSING	System	5 min	Alert after 15 min	Start of OCR/parse

Workflow ID	Workflow Name	Current State	Trigger	Condition	Next State	Actor	SLA Timer	Escalation	Notes
	extraction							waiting	pipeline.
WF-ING-01	Ingestion and extraction	PROCESSING	OCR and schema extraction complete	All mandatory validations pass and confidence >= thresholds	EXTRACTED	System	15 min	Escalate to admin if > 30 min	Successful machine extraction.
WF-ING-01	Ingestion and extraction	PROCESSING	Validation complete	Any field below threshold or issue present	REVIEW_REQUIRED	System	15 min	Notify district queue immediately	Human review required.
WF-ING-01	Ingestion and extraction	REVIEW_REQUIRED	Reviewer accepts/corrects all required fields	No blocking issues remain	APPROVED	District Operator / Reviewer	1 business day	Escalate to supervisory queue after SLA breach	Approved extraction may publish profile/report updates.
WF-ING-01	Ingestion and extraction	ANY_ACTIVE	Processing error	Retry count < limit	RETRY_PENDING	System	Backoff 5/15/60 min	Admin alert on third failure	Transient failure handling.
WF-ING-01	Ingestion and extraction	RETRY_PENDING	Retry exhausted	Retry count >= limit	FAILED	System	After final retry	Admin queue + connector health alert	Manual intervention required.
WF-PROF-01	Subject profile publication	DRAFT	New evidence linked	Candidate subject resolved	PENDING_REVIEW	System	Immediate	None	Profile proposal created.
WF-PROF-01	Subject profile publication	PENDING_REVIEW	Reviewer approves	No unresolved conflicts	PUBLISHED	Reviewer	1 business day	Escalate to district supervisor	Published profile visible in search/reporting.
WF-PROF-01	Subject profile publication	PENDING_REVIEW	Reviewer identifies conflict	Conflict unresolved	CONFLICTING	Reviewer	1 business day	Escalate after 2 business days	Profile remains partial until resolved.
WF-PROF-01	Subject profile publication	CONFLICTING	Conflict resolved	Reviewer decision saved	PUBLISHED	Reviewer	1 business day	Escalate to HQ legal/intelligence if overdue	Conflict history retained.
WF-IR-01	Interrogation report	DRAFT	Generate report	Template active and sources valid	REVIEW_REQUIRED	System	Immediate	None	Generated draft awaiting completion.
WF-IR-01	Interrogation report	REVIEW_REQUIRED	User completes missing fields	All mandatory fields present	APPROVAL_PENDING	District Operator	4 business hours	Escalate to supervisor after SLA breach	Ready for final approval.
WF-IR-01	Interrogation report	APPROVAL_PENDING	Approver approves	Role and jurisdiction valid	FINALIZED	Supervisory Officer / Legal Reviewer	1 business day	Escalate to HQ after SLA breach	Final PDF/DOCX available.
WF-IR-01	Interrogation report	APPROVAL_PENDING	Approver rejects	Reason code mandatory	DRAFT	Supervisory Officer / Legal Reviewer	Immediate	Notify creator	Returned for correction.
WF-LEGAL-01	E-Courts legal update	MONITORING	Court order found	Match confidence >= auto-propose threshold	REVIEW_PENDING	System	Immediate	Notify legal reviewer	Proposed legal update created.
WF-LEGAL-01	E-Courts legal update	REVIEW_PENDING	Reviewer approves	Order valid	APPLIED	Legal Reviewer	1 business day	Escalate to HQ legal queue	Case and profile updated.
WF-LEGAL-01	E-Courts legal update	REVIEW_PENDING	Reviewer rejects	Ambiguous or incorrect match	DISCARDED	Legal Reviewer	1 business day	None	Evidence retained but not applied.
WF-REQ-01	Unocross requisition	DRAFTED	Rule engine triggers	Evidence threshold met	REVIEW_PENDING	System	Immediate	Notify analyst queue	Draft package created.
WF-REQ-01	Unocross requisition	REVIEW_PENDING	Approver approves	Draft content complete	APPROVED	Supervisory Officer	4 business hours	Escalate after SLA breach	Eligible for export/send.
WF-REQ-01	Unocross requisition	APPROVED	Dispatch action invoked	Approved channel available	DISPATCHED	Approver/System	30 min	Notify if dispatch fails	Dispatch audit recorded.
WF-REQ-01	Unocross	REVIEW_PENDING	Approver rejects	Reason required	REJECTED	Supervisory Officer	4 business hours	None	Draft retained for

Workflow ID	Workflow Name	Current State	Trigger	Condition	Next State	Actor	SLA Timer	Escalation	Notes
	requisition								audit.
WF-LEAD-01	Lead to memo routing	NEW	Lead submitted	Required fields valid	VALIDATED	Operator/System	15 min for critical, 1 hour for others	HQ fallback after SLA breach	Lead stored and duplicate check completed.
WF-LEAD-01	Lead to memo routing	VALIDATED	Memo generated	Template available	MEMO_GENERATED	System	15 min	Notify approver	Memo created.
WF-LEAD-01	Lead to memo routing	MEMO_GENERATED	Submit for approval	Routing target resolved	APPROVAL_PENDING	Operator/System	Immediate	None	Waiting approver action.
WF-LEAD-01	Lead to memo routing	APPROVAL_PENDING	Approver approves	Jurisdiction valid	ROUTED	SP / Zonal Officer	30 min critical, 4 hours high, 1 business day others	Escalate to HQ if not acted	Memo routed to target officer.
WF-LEAD-01	Lead to memo routing	ROUTED	Target acknowledges	Owner assigned	IN_ACTION	Receiving Officer	4 business hours	Escalate to supervising officer	Execution underway.
WF-LEAD-01	Lead to memo routing	IN_ACTION	Closure submitted	Closure code and notes present	CLOSED	Receiving Officer / Supervisor	As per case	Escalate overdue open leads weekly	Full digital trail retained.
WF-ALERT-01	Alert handling	OPEN	User acknowledges	Authorized role	ACKNOWLEDGED	Analyst / Officer	5 min critical, 30 min high, 4 hours medium, 1 business day low	Escalate to next level after breach	Acknowledgement starts response tracking.
WF-ALERT-01	Alert handling	ACKNOWLEDGED	Assign owner	Assignee valid	ASSIGNED	Analyst / Supervisor	Immediate	None	Ownership established.
WF-ALERT-01	Alert handling	ASSIGNED	No action before SLA due	SLA breached	ESCALATED	System	As configured by severity	Escalate to supervisory queue	Escalation history recorded.
WF-ALERT-01	Alert handling	ASSIGNED	Resolution submitted	Evidence attached if required	RESOLVED	Assignee	Variable by alert type	Supervisor review if configured	Resolution recorded.
WF-ALERT-01	Alert handling	RESOLVED	Supervisor closes	Closure valid	CLOSED	Supervisor	1 business day	Escalate unresolved closure review	Closed record immutable except admin note.

8. Non-Functional Requirements

NFR ID	Category	Requirement	Verification Method	RFP Ref
NFR-PER-001	Performance	P95 search response for standard indexed queries shall be <= 3 seconds on benchmark production hardware.	Load test with 250 concurrent sessions	RFP-TECH-08
NFR-PER-002	Performance	P95 dossier open time shall be <= 5 seconds when required source enrichments are locally available.	Load test / APM traces	RFP-P2-01
NFR-PER-003	Performance	Interrogation report generation from already-ingested data shall complete within <= 60 seconds for normal cases.	Timed job benchmark	RFP-P1-11
NFR-PER-004	Performance	OCR plus extraction of a 50-page PDF shall complete within <= 120 seconds on provisioned processing nodes.	Worker benchmark	RFP-P1-13
NFR-PER-005	Performance	Priority alert creation from ingested real-time location or scoring event shall occur within <= 5 minutes of event receipt.	Synthetic event test	RFP-BIZ-07
NFR-PER-006	Performance	Tower dump jobs up to 5 million records shall complete within <= 20 minutes on reference batch hardware or surface queued progress if capacity constrained.	Batch benchmark	RFP-P2-05

NFR ID	Category	Requirement	Verification Method	RFP Ref
NFR-SCL-001	Scalability	System shall support 1,000 named users and 250 concurrent interactive sessions.	Capacity test	RFP-TECH-08
NFR-SCL-002	Scalability	System shall support at least 1 million subject profiles, 10 million cases/documents, and 100 million communication events without redesign.	Data volume benchmark	RFP-TECH-08
NFR-SEC-001	Security	All browser and API traffic shall use TLS 1.2 or higher.	Security configuration review	RFP-TECH-01
NFR-SEC-002	Security	Sensitive fields at rest shall be encrypted using AES-256 or Department-approved equivalent; encryption keys shall be managed in approved secrets infrastructure.	Security review / penetration test	RFP-TECH-01
NFR-SEC-003	Security	Field-level masking shall apply to Aadhaar, PAN, bank, passport/visa, and chat evidence unless role-specific permission grants full view.	RBAC test	RFP-TECH-07
NFR-SEC-004	Security	All approvals, exports, dispatches, and evidence access shall be auditable and tamper-evident.	Audit log verification	RFP-BIZ-06
NFR-SEC-005	Security	Outbound network access shall be allow-listed; no public AI inference endpoints may be contacted.	Network security review	RFP-TECH-02
NFR-REL-001	Reliability	Target application uptime shall be 99.5% excluding approved maintenance windows.	Monthly SLA report	RFP-TECH-12
NFR-REL-002	Reliability	A connector failure shall not block other ingestion pipelines or primary UI availability.	Chaos/failure test	RFP-P1-03
NFR-REL-003	Reliability	Backups and disaster recovery shall support RPO <= 15 minutes and RTO <= 4 hours where infrastructure permits.	DR drill	RFP-TECH-11
NFR-OBS-001	Observability	System shall expose health checks, worker queue depth, error rates, connector health, and audit analytics to admins.	Operational monitoring review	RFP-TECH-05
NFR-OBS-002	Observability	All jobs and API requests shall carry correlation IDs across logs, metrics, and audit events.	Trace inspection	RFP-TECH-05
NFR-LOC-001	Localization	UI, OCR, and reports shall support English and Telugu with Unicode storage and transliteration-aware search.	Functional test	RFP-P1-13
NFR-AIQ-001	AI Quality	High-priority extraction fields shall target precision >= 0.90 on the Department-approved validation set before production sign-off.	Model evaluation report	RFP-P1-13
NFR-AIQ-002	AI Quality	Role classification shall target macro-F1 >= 0.80 on the curated benchmark; otherwise the feature shall operate in manual-review or rule-only mode.	Model evaluation report	RFP-P2-07

NFR ID	Category	Requirement	Verification Method	RFP Ref
NFR-AIQ-003	AI Quality	Every AI-generated summary, legal mapping, or score explanation shall cite source evidence and model version.	Governance test	RFP-TECH-03
NFR-ACC-001	Accessibility	Internal web UI shall conform to WCAG 2.1 AA to the extent practical for an intranet application.	Accessibility review	RFP-TECH-08
NFR-MNT-001	Maintainability	Templates, KPI dictionaries, risk rules, routing rules, and geofences shall be editable by authorized admins without code changes.	Admin configuration test	RFP-TECH-08
NFR-MNT-002	Maintainability	All deployment artifacts, configuration files, API contracts, and runbooks shall be version-controlled and deliverable to the Department.	Handover checklist	RFP-TECH-09
NFR-RET-001	Retention	Retention and archive policies shall be configurable by entity type and respect legal hold.	Retention policy test	RFP-BIZ-06

9. Integration Requirements

Integration ID	External System	Purpose	Interface Type	Direction	Data Format	Authentication Method	Frequency	Retry Mechanism	Failure Fallback	RFP Ref
INT-001	CCTNS	Subject photo, case/crime, identity, legal context	API / DB replica / secure connector	Inbound	JSON / relational rows / files	Service account, mTLS, or DB credentials	Scheduled + on-demand lookups	3 retries with exponential backoff; queue failures	Manual upload or cached prior snapshot	RFP-P1-03
INT-002	C-DAT / CDAT	Telecom analysis inputs, link analysis, LBS enrichment	API / DB / file ingestion	Inbound	CSV / JSON / DB rows	Service account / DB credentials	Scheduled + job-based	3 retries; isolate failures by batch	Manual import of CDAT extracts	RFP-P1-03
INT-003	CDR	Call detail records	File / API / DB ingestion	Inbound	CSV / TSV / JSON	SFTP, API key, or DB credentials	Batch and on-demand	3 retries per batch file	Manual file import	RFP-P1-03
INT-004	IPDR	Internet protocol detail records	File / API / DB ingestion	Inbound	CSV / JSON	SFTP, API key, or DB credentials	Batch	3 retries per batch	Manual file import	RFP-P1-03
INT-005	FIR Repository	Primary case document source	File/API/ document store	Inbound	PDF / DOCX / JSON	Service account / API token	Scheduled + manual	3 retries	Manual upload	RFP-P1-03
INT-006	Confession cum Seizure Memos	Case fact extraction source	Watched folder / upload / API	Inbound	PDF / JPG / PNG / DOCX	Folder credential / upload auth	Manual + scheduled	3 retries if connector-based	Manual upload	RFP-P1-03
INT-007	IR Repository	Historical IR lookup and enrichment	File / API	Inbound	PDF / DOCX / JSON	Service account / API token	Scheduled + on-demand	3 retries	Manual upload	RFP-P1-03
INT-008	eSakshya	Evidence/ document source	API / file sync	Inbound	JSON / PDF / media refs	Service account / API token	Scheduled + on-demand	3 retries	Manual evidence upload	RFP-P1-03
INT-009	C-Trace	OSINT enrichment source	Approved connector / import feed	Inbound	JSON / CSV / URL refs	API token / file bridge	Scheduled + analyst-triggered	3 retries	Manual import	RFP-P1-03
INT-010	Google Drive / approved equivalent	Monthly Report storage source	OAuth / service account / file bridge	Inbound	PDF / XLSX	OAuth service account or approved local bridge	Scheduled sync	Retry sync 3 times	Manual MR upload	RFP-P1-05
INT-011	E-Courts	Judgment and bail order monitoring	Browser automation / scraping / approved API	Inbound	HTML / PDF / metadata	Approved connector credentials if any	Periodic monitoring	Retry page fetch; mark unavailable after threshold	Manual upload of court orders	RFP-P1-08
INT-012	Internal SMTP / email relay	Dispatch of approved memos and requisitions	SMTP / secure mail relay	Outbound	Email text + attachments	Named sender identity or service account	Event-driven	Retry 3 times; then failed-dispatch queue	Manual download/print/email outside system	RFP-P2-10
INT-013	TS-COP / biometric service	Face-match search where approved	API	Inbound/on-demand	Image / JSON	Service account + legal approval	On-demand only	2 retries; disable after failure threshold	Hide feature if unavailable	RFP-P2-01
INT-014	ICJS	Cross-domain enrichment	API / DB link / file	Inbound	JSON / relational rows	Approved credentials	Feature-flagged batch/on-demand	3 retries	Disabled until approved	RFP-P1-04
INT-015	Transport databases	Vehicle and DL enrichment	API / DB link / file	Inbound	JSON / relational rows	Approved credentials	Feature-flagged	3 retries	Disabled until approved	RFP-P1-04
INT-016	e-Prisons	Prison linkage enrichment	API / DB link / file	Inbound	JSON / relational rows	Approved credentials	Feature-flagged	3 retries	Disabled until approved	RFP-P1-04
INT-017	IVFRT	Immigration / passport enrichment	API / DB link / file	Inbound	JSON / relational rows	Approved credentials	Feature-flagged	3 retries	Disabled until approved	RFP-P1-04
INT-018	NHAI	Toll/vehicle movement enrichment	API / file	Inbound	JSON / CSV	Approved credentials	Feature-flagged	3 retries	Disabled until approved	RFP-P1-04
INT-019	Land/Stamp records	Property and document enrichment	API / file	Inbound	JSON / CSV / PDF refs	Approved credentials	Feature-flagged	3 retries	Disabled until approved	RFP-P1-04
INT-020	Neighbouring -state APIs	Cross-state intelligence enrichment	API / file	Inbound	JSON / CSV	Approved credentials	Feature-flagged	3 retries	Disabled until approved	RFP-P1-04

10. Compliance & Regulatory Mapping

This section captures system controls and audit artefacts derived from the tender. It is not legal advice; it is an implementation control map.

Compliance ID	Compliance Requirement	Mapped Functional Requirement(s)	System Control	Audit Artifact	RFP Ref
CMP-001	On-prem data residency and model sovereignty	FR-01, FR-02, FR-26; NFR-SEC-005; INT-001..020	No public AI APIs; all data/model artifacts stored on Department infrastructure; egress allow-list only	Deployment diagrams, firewall rules, model registry export	RFP-TECH-01, RFP-TECH-02
CMP-002	Role-based access and least privilege	FR-01; UI section 11; Audit section 13	Role-action matrix, jurisdiction scoping, server-side authorization, masked-field policy	User-role matrix, authorization test evidence, access logs	RFP-TECH-07
CMP-003	Human approval before external legal or financial communication	FR-07, FR-16, FR-21, FR-24	Approval workflows with named approver, maker-checker option, immutable audit trail	Approval logs, dispatch logs, memo/requisition versions	RFP-P1-10, RFP-P2-10, RFP-BIZ-05
CMP-004	Digital evidence integrity and chain-of-custody	FR-22; Data section 5; Audit section 13	SHA-256 hashing, immutable object storage, legal hold, integrity verification API	Hash verification reports, evidence access/export logs, legal hold records	RFP-BIZ-06
CMP-005	Traceable legal status updates	FR-06, FR-21	Store court order metadata, review decisions, prior status history, legal mapping version	Legal review queue records, order metadata, audit history	RFP-P1-08, RFP-BIZ-05
CMP-006	Privacy of sensitive identifiers and financial/chat evidence	FR-01, FR-22; NFR-SEC-002, NFR-SEC-003	Field-level masking, encryption at rest, restricted export, redacted reporting	Field access logs, penetration test results, masked export samples	RFP-P1-01
CMP-007	Multilingual processing and standardization	FR-03, FR-08; NFR-LOC-001	Telugu and English OCR, transliteration-aware search, bilingual templates where required	OCR benchmark results, bilingual UAT cases	RFP-P1-13
CMP-008	Explainable AI and reviewable decision support	FR-20, FR-21, FR-26; NFR-AIQ-003	Evidence-cited outputs, factor breakdown, model version tracking, manual override	Model evaluation report, review decisions, prompt/version registry	RFP-BIZ-03, RFP-BIZ-04, RFP-TECH-03
CMP-009	Operational SLA and escalation governance	FR-24; Workflow section 7; Audit section 13	SLA timers, escalation rules, acknowledgement tracking, overdue reporting	Alert aging report, workflow task logs, escalation metrics	RFP-BIZ-07
CMP-010	Department ownership and customizability	Section 16 Delivery/Handover; FR-23; Implementation guidance	Source code, configs, prompts, model artifacts, admin access, documentation handover	Handover checklist, repository export, admin credential transfer record	RFP-TECH-10
CMP-011	Warranty, support, and training obligations	Section 16 Delivery/Handover	Support staffing, training sessions, hypercare plan, warranty SLA matrix	Attendance sheets, support roster, issue logs, acceptance sign-offs	RFP-TECH-05, RFP-TECH-12, RFP-TECH-13
CMP-012	Contractual governance and auditability	Section 16 Delivery/Handover; Appendices A/B	Capture payment-linked acceptance, penalties, PBG references, and procurement constraints for governance traceability	Acceptance milestones, risk register, contractual appendix	RFP-CON-01..07

11. UI / UX Requirements

Screen ID	Screen Name	Role Visibility	Input Fields / Main Controls	Validation Behaviour	Error Display Rules	Conditional Rendering Logic
SCR-01	Login / SSO	All users	Username/SSO button, optional MFA, language selector	SSO required if configured; local login only for approved break-glass accounts	Inline field errors; auth failures shown without revealing account existence	Hide local login when SSO-only mode enabled
SCR-02	Home Dashboard	Role-specific	Global filters, KPI cards, recent alerts, task queue, stale-data indicators	Date range default to last 30 days; filters limited by jurisdiction	Errors shown as widget-level banners; dashboard still renders partial content	Widgets hidden if role lacks permission or source data disabled
SCR-03	Ingestion Queue	District Operator, HQ Admin, System Admin	Source, document type, file name,	Manual upload requires document	Per-row failure reason and retry	Retry action visible only to permitted

Screen ID	Screen Name	Role Visibility	Input Fields / Main Controls	Validation Behaviour	Error Display Rules	Conditional Rendering Logic
			status, retry count, issues	type and source selection	action	roles and failed states
SCR-04	Document Review	District Operator, Legal Reviewer, Analyst	Document viewer, extracted fields, confidence, source spans, review actions	Reviewer value required for corrected fields; mandatory fields cannot be left empty when finalizing	Field-level validation messages and page-level issue summary	Source pane collapses on smaller screens; legal fields visible only to authorized roles
SCR-05	Subject Profile	IO, Analyst, Legal Reviewer, HQ Admin	54-column sections, completeness score, evidence links, conflicts, history	Masked fields only revealed to permitted roles; conflict resolution requires reason	Section-level error if related source unavailable	Edit actions hidden when profile status is published and user lacks unlock permission
SCR-06	Search and Dossier	Analyst, IO, Supervisory Officer	Search bar, query type, filters, result grid, match reasons, export	Sensitive query types require permission; page size max 100	No-result state with suggestions; backend errors shown as search banner	Biometric tile hidden when connector disabled
SCR-07	Network Analysis Graph	Analyst, IO	Seed selector, depth, date range, filters, graph canvas, ranked nodes	Depth max from config; at least one seed required	Async job failure shown with retry option	Export visible only after job completion
SCR-08	Technical Analysis Report	IO, Analyst	Subject/number selector, time window, report preview, export buttons	Time window required; LBS-only sections hidden if data absent	Section-level unavailable labels, not blank spaces	Route maps hidden when map service unavailable but tables still shown
SCR-09	Monthly Report Dashboard	HQ Admin, Analyst, Auditor	Reporting month, district filter, KPI grid, trend charts, review queue link	District-month uniqueness enforced on approval	KPI parsing issues linked to review queue	Approval controls hidden for read-only roles
SCR-10	Alerts Center	Analyst, Supervisor, HQ Admin	Alert list, severity, type, assignee, SLA timer, acknowledge/escalate actions	Critical alerts require immediate action; close requires closure code	Toast + row error on failed action	Escalate/close buttons visible only for PL-3 or configured roles
SCR-11	Lead Entry	Toll-Free Operator, District Operator	Informant fields, channel, location, suspect details, urgency, attachments	Channel, location, urgency mandatory; duplicate warning shown before submit	Field-level validation and duplicate candidate modal	Anonymous lead fields rendered when operator selects anonymous flag
SCR-12	Memo / Requisition Approval Queue	Supervisor, Legal Reviewer, HQ Admin	Pending items, template preview, evidence summary, approve/reject/dispatch	Approve requires decision and, where configured, dispatch channel	Action failure displayed inline and logged	Dispatch controls visible only after approval or when auto-dispatch disabled
SCR-13	Legal Review Queue	Legal Reviewer, HQ Admin	Court orders, candidate sections, match confidence, approve/reject actions	Decision requires reason when rejecting or modifying	Ambiguous matches highlighted with warning badge	Auto-apply controls hidden because human approval mandatory
SCR-14	MIS Report Builder	HQ Admin, Analyst, Auditor	Report catalog, filters, schedule, format, preview, export	Only authorized reports/formats visible	Export job errors displayed in job drawer	Scheduling controls hidden for read-only auditor
SCR-15	Admin Configuration	HQ Admin, System Admin	Templates, KPI dictionary, routing rules, geofences, feature flags, risk weights	Schema validation before save; activation requires approval if configured	Validation errors shown with exact field path	Feature sections hidden based on admin role domain
SCR-16	Audit Trail Viewer	Auditor, HQ Admin, IT/Security	Date range, actor, entity, action, export, integrity view	Date range mandatory; export limited by policy	Query failure shown without exposing internal stack traces	Sensitive snapshot fields masked unless security role permits

12. Reporting & Analytics

Report ID	Report Name	Data Source	Filters	Aggregations / Output	Export Formats
REP-001	Statewide Monthly KPI Dashboard	MR KPI tables	Reporting month, district, zone	Trend lines, totals, district ranking	PDF, XLSX
REP-002	District Operational Performance Report	Cases, leads, alerts, MR KPIs	District, date range, drug type	Counts, averages, closure rates	PDF, XLSX
REP-003	Subject Dossier Export	Subject profile, cases, alerts, evidence	Subject ID, date range	Narrative summary + linked tables	PDF
REP-004	Interrogation Report	FIR, extracted fields, supplements	Case ID, template version	Template-based document	PDF, DOCX

Report ID	Report Name	Data Source	Filters	Aggregations / Output	Export Formats
REP-005	Technical Analysis Report	CDR/CDAT/LBS, device history	Subject/number, date range	Top contacts, routes, stay analysis	PDF
REP-006	Network Analysis Report	Graph nodes/edges, cases, communications	Seed set, depth, date range	Ranked nodes, bridge counts, graph stats	PDF, PNG, XLSX
REP-007	Lead Aging and Routing Report	Lead, memo, workflow task	District, urgency, status, age bucket	Open counts, SLA breaches, closure trends	PDF, XLSX
REP-008	Alert Operations Report	Alert, workflow task	Severity, type, district, date range	Alert volume, acknowledgement time, escalation rates	PDF, XLSX
REP-009	Legal Status Change Report	Legal orders, cases, subjects	Date range, district, court, order type	New bail orders, judgments, status changes	PDF, XLSX
REP-010	Audit and Export Activity Report	Audit log	Date range, actor, action, target type	Counts, exception events, export history	PDF, XLSX
REP-011	Model Quality and Governance Report	ModelVersion, evaluations, overrides	Model, feature, date range	Benchmark metrics, override rates, drift indicators	PDF, XLSX
REP-012	Connector Health and Data Freshness Report	ConnectorConfig, ingestion jobs	Source, date range, status	Success rate, last sync, stale sources	PDF, XLSX

13. Audit & Logging

Log ID	Log Category	What Is Logged	Who Can View	Retention Period	Notes
LOG-001	Authentication events	Login success/failure, MFA challenge, logout, session expiry	IT/Security, System Admin, Auditor	365 days	Mask secrets; include source IP and auth provider when available
LOG-002	Authorization failures	Forbidden actions, restricted field access attempts	IT/Security, Auditor	365 days	High-value for security monitoring
LOG-003	Record create/update/delete-intent events	Entity changes, review decisions, merge operations	HQ Admin, Auditor, authorized supervisors	7 years or legal hold	Hard deletes prohibited for evidence-bearing entities
LOG-004	Approval and dispatch events	Memo approval, requisition approval, dispatch, rejection	Auditor, HQ Admin, Legal Reviewer	7 years	Named approver mandatory
LOG-005	Evidence access and export events	View, download, print, watermark export, hash verify	Auditor, IT/Security, HQ Admin	7 years or legal hold	Include justification code when required
LOG-006	AI interaction metadata	Prompt template ID, model version, token counts, response status, citation count	Model Governance Reviewer, IT/Security, Auditor	3 years	Sensitive prompt/response content visible only to authorized reviewers
LOG-007	Connector and ingestion job events	Sync start/end, retries, failures, schema changes	System Admin, HQ Admin	180 days	Used for operational support and freshness monitoring
LOG-008	Alert and SLA events	Creation, acknowledgement, escalation, closure, breach	Supervisor, HQ Admin, Auditor	5 years	Supports operational accountability
LOG-009	Configuration and feature flag changes	Template activation, rule changes, geofence changes, user-role changes	HQ Admin, IT/Security, Auditor	7 years	Maker-checker approval trail required for governed configs
LOG-010	System health and security events	Service restart, backup, restore, integrity failure, suspicious activity	IT/Security, System Admin	1 year	Critical security events should trigger alerts

14. Test Case Derivation Section

Test Case ID	FR ID	Test Scenario	Expected Result
TC-FR-01-01	FR-01	Attempt to export a masked dossier using a role without export permission.	System denies the action, keeps masked fields hidden, and writes an authorization-failure audit event.
TC-FR-02-01	FR-02	Submit a valid FIR PDF upload and create an ingestion job.	System stores the original file, creates a unique job, and moves the job through queued processing states.
TC-FR-03-01	FR-03	Upload a bilingual Telugu/English Panchanama with one low-confidence field.	System extracts fields, flags the low-confidence item for review, and preserves field-level provenance.

Test Case ID	FR ID	Test Scenario	Expected Result
TC-FR-04-01	FR-04	Ingest new evidence for an existing subject where one new value conflicts with an approved high-trust value.	System keeps both assertions, marks the field CONFLICTING, and does not overwrite the approved value automatically.
TC-FR-05-01	FR-05	Sync a district Monthly Report PDF containing all 20 KPIs.	System parses KPIs, routes no-review-needed metrics to approval, and publishes final values to reporting after approval.
TC-FR-06-01	FR-06	Find a new bail order in E-Courts for a monitored case.	System creates a proposed legal update with court metadata and routes it to legal review before applying.
TC-FR-07-01	FR-07	Import transaction data showing repeated transfers to the same counterparty with a matched telecom identity.	System generates one approval-pending Unocross draft with evidence summary and suppression-safe idempotent behavior.
TC-FR-08-01	FR-08	Generate an interrogation report draft where two mandatory fields are unavailable from source data.	System creates the draft, flags the missing fields in manual-entry form, and blocks finalization until resolved.
TC-FR-09-01	FR-09	Search for a subject by alias transliterated from Telugu to English.	System returns ranked matches with match reasons and allows dossier opening for permitted users.
TC-FR-10-01	FR-10	Ask the assistant for repeat offenders with new bail orders in the last 30 days.	System returns a grounded summary with cited records, applied filters, and model version metadata.
TC-FR-11-01	FR-11	Launch a depth-3 network job from a seed subject.	System runs the job asynchronously, ranks common nodes, and exposes graph/table results with factor explanation.
TC-FR-12-01	FR-12	Generate a technical analysis report for a number with missing route data but available contact data.	System renders available sections, marks route section as Data unavailable, and still produces a valid report.
TC-FR-13-01	FR-13	Feed a watchlisted target location event entering the Goa geofence.	System creates a geofence alert with correct severity, starts SLA timer, and suppresses duplicates within the configured window.
TC-FR-14-01	FR-14	Upload a large tower dump file with mixed-format phone numbers.	System normalizes valid numbers, reports invalid entries separately, ranks matched results, and preserves job audit history.
TC-FR-15-01	FR-15	Run role classification on a case aligned to a known peddler pattern from the benchmark set.	System returns PEDDLER as candidate role with confidence, evidence snippets, and review status.
TC-FR-16-01	FR-16	Create a high-urgency WhatsApp lead that matches an existing open lead.	System warns of duplicate candidate, allows justified override, creates the lead, and routes memo generation workflow.
TC-FR-17-01	FR-17	Schedule a district operational report to PDF and XLSX.	System creates export jobs, applies selected filters, watermarks outputs, and records export audit entries.
TC-FR-18-01	FR-18	Enable a new optional Transport connector in test mode.	System validates the connector config, keeps it feature-flagged until approval, and isolates any failures from core pipelines.
TC-FR-19-01	FR-19	Ingest a public social-media screenshot and matching text export from an approved source.	System stores both as evidence, clusters them as one content item, and preserves separate provenance records.
TC-FR-20-01	FR-20	Score a monitored content item linked to a watchlisted subject and containing high-risk keywords.	System assigns a controlled category, calculates a 0-100 risk score with factor breakdown, and routes the item to the priority queue.
TC-FR-21-01	FR-21	Generate legal section suggestions from FIR and confession facts.	System proposes candidate sections with source-backed rationale and blocks final application until reviewer approval.
TC-FR-22-01	FR-22	Run integrity verification on an evidence file after retrieval.	System recalculates the SHA-256 hash, confirms match, and logs the verification event; mismatch would quarantine the artifact.
TC-FR-23-01	FR-23	Activate a new memo template version with maker-checker enabled.	System saves draft version, requires approval, activates only after approver sign-off, and keeps prior version available for history.
TC-FR-24-01	FR-24	Leave a critical alert unacknowledged past its SLA threshold.	System escalates the alert to the configured supervisory role and logs the escalation timestamp and owner history.
TC-FR-25-01	FR-25	Merge two duplicate subject candidates with overlapping cases and documents.	System creates one surviving subject, preserves lineage, updates linked records, and writes a merge audit trail.

Test Case ID	FR ID	Test Scenario	Expected Result
TC-FR-26-01	FR-26	Attempt to promote a model version whose benchmark score is below threshold.	System rejects production promotion and places the dependent feature into manual-review or rule-only mode.

15. Implementation Guidance for AI Development

15.1 Recommended Service Boundaries

Delivery strategy recommendation: because the tender contains a 6-week delivery expectation, Release 1 should be implemented as a modular monolith with isolated worker processes rather than a fully distributed microservice estate. Service boundaries below should be treated as module boundaries first and extraction-to-microservice candidates later.

Service ID	Suggested Service / Module	Responsibilities	Primary Data Store / Schema	Primary API Surface
SRV-01	Identity and Access	User auth broker integration, roles, permissions, jurisdiction scope, session policy, audit hooks	identity_access schema, cache	/auth, /users, /roles
SRV-02	Ingestion and Evidence	File intake, connector orchestration, immutable storage, checksums, legal hold, evidence retrieval	evidence_ingestion schema, object storage	/ingestion-jobs, /documents, /evidence
SRV-03	Extraction and NLP	OCR, extraction, bilingual parsing, field review queues	extraction schema, model registry	/extractions, /ocr, /reviews
SRV-04	Subject and Case Intelligence	Subject resolution, profile publication, case linkage, legal section suggestions	case_intelligence schema, search index	/subjects, /cases, /legal-mapping
SRV-05	Analytics and Graph	Search, dossier assembly, link analysis, tower dump ranking, technical report data prep	analytics schema, graph store/search index	/search, /network-analysis, /technical-analysis
SRV-06	Workflow and Approvals	Tasks, approvals, routing, SLA timers, notifications, escalations	workflow schema	/tasks, /approvals, /alerts, /notifications
SRV-07	Lead and Memo Management	Lead intake, memo generation, routing to officers	lead_mgmt schema, template store	/leads, /memos
SRV-08	Reporting and Exports	MIS definitions, dashboards, exports, scheduled jobs	reporting schema, export store	/reports, /exports, /dashboards
SRV-09	AI Runtime and Governance	Model serving, prompt registry, validation, approvals, scoring, categorization	ai_governance schema, model storage	/ai/query, /models, /benchmarks
SRV-10	Admin Configuration	Templates, KPI dictionary, feature flags, geofences, routing rules	admin_config schema	/admin/config, /templates

15.2 Suggested DB Schema Groupings

Group ID	Schema / Domain	Suggested Tables
DB-GRP-01	identity_access	users, roles, permissions, sessions, jurisdiction scopes
DB-GRP-02	evidence_ingestion	ingestion_jobs, source_documents, source_payloads, chain_of_custody, storage_refs
DB-GRP-03	extraction	extraction_fields, extraction_versions, review_decisions, ocr_pages
DB-GRP-04	case_intelligence	subject_profiles, subject_aliases, subject_assertions, cases, legal_orders, subject_case_links
DB-GRP-05	telecom_finance	communication_events, device_history, financial_transactions, location_events
DB-GRP-06	network_analytics	network_nodes, network_edges, graph_jobs, tower_dump_jobs, ranking_results
DB-GRP-07	workflow_ops	workflow_tasks, alerts, notifications, approvals, escalations, SLA policies
DB-GRP-08	lead_mgmt	leads, memos, routing_rules, channel_sources

Group ID	Schema / Domain	Suggested Tables
DB-GRP-09	reporting	report_definitions, schedules, report_runs, KPI_fact tables, dashboard caches
DB-GRP-10	ai_governance	model_versions, prompt_templates, benchmark_runs, override_logs
DB-GRP-11	admin_config	master_data, geofences, template_versions, feature_flags, connector_configs

15.3 Suggested Frontend Module Breakdown

Frontend Module ID	Module Name	Scope
FE-01	Shell and Access	App shell, SSO, session management, language switch, role-aware navigation
FE-02	Ingestion Workspace	Upload flows, queue views, retry controls, evidence metadata
FE-03	Document Review	Viewer, OCR overlays, extraction grid, comparison and review actions
FE-04	Profile and Case Workspace	Subject profile, case view, conflict resolution, legal suggestions
FE-05	Search and Dossier	Universal search, result grids, dossier tabs, export
FE-06	Analytics Workspace	Graph view, technical analysis, tower dump, maps
FE-07	Operations Workspace	Alerts center, leads, memo approvals, legal review queue
FE-08	Reporting Workspace	Dashboards, report builder, export jobs, schedules
FE-09	Administration Workspace	Templates, rules, geofences, users, connectors, model governance

15.4 Suggested Folder Structure

```
repo/
 apps/
  web/
  admin-console/
 services/
  api/
  worker/
  ai-runtime/
 packages/
  contracts/
   openapi/
   json-schemas/
  ui/
  config/
  prompts/
 db/
  migrations/
  seeds/
  views/
 infra/
  docker/
```

```
    kubernetes/
    ansible/
    monitoring/
  tests/
    contract/
    integration/
    e2e/
    benchmark/
  docs/
    architecture/
    runbooks/
    training/
```

15.5 Suggested API Grouping

API Group ID	Grouping	Endpoints
API-GRP-01	Identity and user context	/auth, /users, /roles, /permissions
API-GRP-02	Evidence and ingestion	/ingestion-jobs, /documents, /evidence
API-GRP-03	Extraction and review	/ocr, /extractions, /reviews
API-GRP-04	Subject and case	/subjects, /cases, /legal-mapping
API-GRP-05	Search and analytics	/search, /dossiers, /network-analysis, /technical-analysis, /tower-dumps
API-GRP-06	Lead, memo, and alert workflow	/leads, /memos, /alerts, /tasks, /notifications
API-GRP-07	Reporting and exports	/reports, /exports, /dashboards
API-GRP-08	AI runtime and governance	/ai/query, /models, /benchmarks, /prompts
API-GRP-09	Admin configuration	/admin/config, /templates, /master-data, /connectors

15.6 Prompt Engineering Notes

- PMPT-01: Use strict JSON-schema output for OCR extraction, legal mapping, scoring explanation, and report population tasks. Reject any response that does not validate.
- PMPT-02: Prompts must require evidence citation using internal document IDs/page spans and must explicitly instruct the model to answer 'INSUFFICIENT_EVIDENCE' when unsupported.
- PMPT-03: Separate retrieval, reasoning, and rendering prompts. Do not allow direct free-form generation for final governed documents without schema validation and human review.
- PMPT-04: Version prompts independently from model binaries; a prompt change requires governance approval and regression tests.
- PMPT-05: Use small task-specific prompts for classification and field extraction; reserve larger summarization prompts for dossier and analytic narrative generation.
- PMPT-06: Persist all prompt templates, benchmark cases, and expected outputs in version control so AI development tools can regenerate and test reliably.

15.7 AI-Assist Development Handoff Guidance

- Generate API contracts first (OpenAPI plus JSON Schemas), then derive backend controllers, typed clients, and form validators from the same source of truth.
- Represent every workflow state machine as configuration or typed enums before building screens.
- Build evidence preservation, audit, and authorization as first-class cross-cutting concerns before advanced analytics.
- Use deterministic seed data for district, police station, roles, document types, and workflow states to stabilize AI-generated code.
- Encode acceptance criteria into automated contract tests and end-to-end tests before UI assembly for governed features.

16. Delivery, Support, Warranty & Acceptance

16.1 Proposed Delivery Milestones

Important: the tender timeline and the BRD phase durations are inconsistent. The milestone plan below is the proposed reconciliation model and requires sponsor confirmation.

Milestone ID	Milestone	Target	Entry / Exit Criteria	Acceptance Artifact
MIL-001	Kickoff and dependency confirmation	Within 5 business days of PO	Confirm scope, source owners, infra readiness, sample data, acceptance plan	Signed inception report
MIL-002	Environment readiness	By Day 10 from PO	Servers, OS, DB, object storage, search, secrets, network, SMTP, connector access	Environment readiness checklist
MIL-003	Base platform deployment	By end of Week 4	Deploy auth, ingestion skeleton, evidence storage, admin config, core UI shell	Deployment verification report
MIL-004	Contractual installation complete	By end of Week 7 including installation week	Production deployment installed on Department infrastructure and smoke tested	Installation sign-off
MIL-005	Phase 1 functional completion	Target Month 4 subject to source access and approved templates	Core data, legal intelligence, reporting, governance features live	Phase 1 UAT sign-off
MIL-006	Phase 2 functional completion	Target Month 8 subject to advanced source availability	Dossier, graph, technical analysis, lead workflow, advanced intelligence live	Phase 2 UAT sign-off
MIL-007	Hypercare	2 weeks post go-live	3-4 support staff on site, daily issue review, knowledge transfer	Hypercare closure report

16.2 Hosting Infrastructure Bill of Requirements

Infrastructure ID	Component	Baseline Requirement
INF-001	Application nodes	2+ CPU application nodes, 8 vCPU / 32 GB RAM each minimum for HA baseline
INF-002	Worker/OCR nodes	2+ worker nodes sized for OCR/extraction throughput; add GPU if model benchmarks require it
INF-003	Database	PostgreSQL HA-capable deployment with SSD-backed storage and encrypted backups
INF-004	Search index	OpenSearch/Elasticsearch cluster sized for full-text search and aggregations
INF-005	Object storage	MinIO or approved equivalent with versioning, immutability/retention support, and encryption
INF-006	Optional graph store	Neo4j or approved graph capability if graph workload exceeds relational/search approach
INF-007	Secrets and certificates	Approved secrets storage plus TLS certificate management
INF-008	Network	Restricted internal network; allow-list egress only for approved sources such as E-Courts and storage bridge
INF-009	Observability	Prometheus/Grafana/ELK or approved equivalent for logs, metrics, traces, and alerts
INF-010	Backup/DR	Snapshot and backup tooling to meet RPO/RTO targets; off-host copy required

16.3 Warranty, Support, Training, and ATS

Support ID	Service	Duration / Trigger	Scope	Acceptance / SLA
SUP-001	Warranty	1 year from production acceptance	Bug fixes, security patches, maintenance releases, approved upgrades, performance tuning	Response SLA: Critical 4h, High 1 business day, Medium 2 business days, Low 5 business days
SUP-002	On-site engineering support	6-8 months as per tender	Implementation support, connector tuning, model tuning, environment assistance, issue triage	Department to confirm exact staffing and working hours
SUP-003	Hypercare staffing	2 weeks post deployment	3-4 technical manpower on site, daily stand-up, rapid incident handling	Daily ticket/status report
SUP-004	Training	Before go-live and during hypercare	Admin training, operator training, analyst training, reviewer training, security/audit training	Attendance sheets, training materials, recorded demos where allowed
SUP-005	ATS transition	Post warranty, if invoked	Support handover, renewal scope, unresolved issues list, config/model baseline snapshot	ATS proposal and handover pack

16.4 Department Ownership and Handover Artefacts

- HO-001: Source code repositories and branch structure
- HO-002: Deployment manifests, infrastructure prerequisites, and environment variables template
- HO-003: Database schema, migration scripts, seed master data, and API contracts
- HO-004: Prompt templates, model registry export, benchmark datasets references, and evaluation reports
- HO-005: Admin manuals, operator manuals, runbooks, backup/restore SOPs, and security hardening guide
- HO-006: Template package, KPI dictionary, routing rules, geofences, feature flag inventory, and active config export
- HO-007: Test evidence, UAT sign-offs, known issues register, and support contact matrix

16.5 ATS Transition Baseline

- Export final configuration baseline, active templates, role mappings, prompt registry, model registry, connector inventory, issue backlog, and known workarounds.
- Provide support runbook, escalation contacts, backup/restore SOP, and open-risk register.
- Freeze and sign off production schema and API versions before ATS handover.

16.6 Non-System Contractual Clauses Preserved for Governance Traceability

Appendix Item	Clause Topic	Tender Requirement
CON-APP-001	Payment milestones	80% on delivery and successful installation; 20% after satisfactory performance report 60 days post installation.
CON-APP-002	Penalty/LD	1% for first week delay, 1.5% for second week, 2% for third week, continuing up to 10%; order cancellation rights after >30 days delay.
CON-APP-003	Performance Bank Guarantee	5% of project value, valid beyond 30 days post warranty period.
CON-APP-004	Warranty baseline	Minimum 1 year comprehensive onsite warranty on entire solution.
CON-APP-005	Consortium/subcontracting	Not permitted under the tender.
CON-APP-006	Governing law and jurisdiction	Indian law; Hyderabad, Telangana jurisdiction; force majeure and notice clauses apply.

Appendix A - Gap Analysis Matrix

RFP Clause	Clause Summary	Present in Current BRD? (Y/N)	Gap Description	Required Enhancement
RFP-BIZ-01	Business objective: AI-powered intelligence platform for narcotics trafficking and digital drug promotion networks	Y	BRD states AI-assisted intelligence platform purpose but does not explicitly call out digital drug promotion monitoring as a distinct operational objective.	Add explicit objective, operating context, and measurable outcomes for narcotics and digital promotion network monitoring.
RFP-BIZ-02	Cross-platform monitoring	N	Current BRD has OSINT/C-Trace inputs but no dedicated requirement for cross-platform monitoring across approved online/public channels.	Add a functional requirement for cross-platform content ingestion, normalization, deduplication, lawful source controls, and monitoring queues.
RFP-BIZ-03	AI-based content categorization	N	No implementable requirement defines category taxonomy, model outputs, review flow, or auditability for AI content categorization.	Add a content categorization service with controlled labels, confidence score, explanation, and manual override.
RFP-BIZ-04	Risk scoring and prioritization	N	No scoring model, scoring factors, thresholds, or prioritization workflow is defined.	Add a configurable risk scoring model with 0-100 scale, weighted factors, threshold-based queues, and review/explanation rules.
RFP-BIZ-05	Legal section mapping	N	BRD stores section_of_law but does not define how facts map to candidate legal sections or how reviewers validate them.	Add legal section mapping requirement with rule/model outputs, rationale, review, and versioned legal mapping rules.
RFP-BIZ-06	Digital evidence preservation	Y	BRD includes provenance and retention but does not fully define chain-of-custody events, integrity verification, legal hold actions, or export controls.	Add digital evidence preservation and chain-of-custody controls including SHA-256 hashing, access ledger, legal hold, and verification APIs.
RFP-BIZ-07	Real-time alerts and escalation	Y	BRD covers geofence alerts only; it does not define a generic alerting and escalation engine, SLA timers, notification channels, or escalation levels.	Add alert/escalation framework with statuses, SLA timers, acknowledgement, reassignment, and escalation rules.
RFP-P1-01	54-column full crime history of a person	Y	Present, but mandatory flags, data dictionary precision, validation rules, and survivorship logic are incomplete.	Expand 54-column schema with attribute-level types, validation rules, provenance, and duplicate/merge rules.
RFP-P1-02	Automatic or semi-automatic population of 54-column schema	Y	Present, but auto-accept thresholds, reviewer gates, and source precedence are not fully specified.	Add confidence thresholds, source trust ranking, approval gates, and field-level conflict handling.
RFP-P1-03	Phase 1 ingestion from CCTNS, C-DAT, CDR, IPDR, FIR, Confession cum Seizure Memo, IR, C-Trace, eSakshya, Google Drive	Y	Present at high level, but per-source interface methods, auth, schedules, retries, and fallback behavior are missing.	Add an integration matrix with method, auth, data formats, sync cadence, retry, and fallback by source.
RFP-P1-04	Phase 3 optional ingestion from ICJS, Transport, e-Prisons, IVFRT, NHAI, Land/Stamp	Y	Present as optional scope, but feature flags, dependency gating, and canonical mapping are not fully specified.	Add pluggable connector framework, approval gates, and mapping rules for optional sources.
RFP-P1-05	Monthly Report ingestion from Google Drive or similar storage	Y	Present, but folder conventions, versioning, and duplicate-finalization controls are not precise enough.	Add naming rules, checksum-based versioning, district-month uniqueness, and review queue criteria.
RFP-P1-06	Extract 20 key MR parameters and update central database	Y	Present, but final KPI dictionary and validation constraints are not defined.	Add KPI master table with metric IDs, formulas, allowed value ranges, and source references.
RFP-P1-07	Automated pictorial and statistical master reports	Y	Present, but report catalog, filters, scheduling rules, and export governance are not specified.	Add reporting catalog with report IDs, filter rules, schedules, and export formats.
RFP-P1-08	Iterative E-Courts scraping for judgments and bail orders	Y	Present, but search keys, rescan cadence, ambiguity handling thresholds, and legal provenance details need precision.	Define E-Courts search strategy, match-confidence thresholds, review workflow, and retention of retrieved orders.
RFP-P1-09	Transaction-CDR cross-check	Y	Present, but data matching keys, confidence rules, and false-link handling are not explicit.	Specify normalization, matching logic, confidence scoring, and partial-match handling.
RFP-P1-10	Automatic draft email/request to Unocross for bank statements	Y	Present, but dispatch controls, template schema, approver roles, and suppression rules need detail.	Add requisition draft template, approval workflow, export/send rules, and suppression/idempotency logic.
RFP-P1-11	Automated interrogation report generation	Y	Present, but versioning, structured field map, and finalization rules need more implementation detail.	Add report template metadata, mandatory field list, role-based editing rules, and export controls.
RFP-P1-12	Mandatory fixed templates for interrogation report	Y	Present, but template governance and effective-dating are not specified.	Add centrally managed template versioning with activation, rollback, and approval workflow.
RFP-P1-13	Panchanama ingestion in text/image/OCR Telugu and	Y	Present, but OCR confidence thresholds, bilingual validation	Add OCR threshold rules, bilingual extraction support, manual

RFP Clause	Clause Summary	Present in Current BRD? (Y/N)	Gap Description	Required Enhancement
	English with missing-field manual entry		rules, and unresolved field status handling are not explicit.	completion workflow, and unresolved field states.
RFP-P2-01	One-click dossier by name/alias/surname/phone/biometric against CCTNS/TS-COP	Y	BRD includes search and optional face match, but TS-COP is not explicitly identified and biometric governance is under-specified.	Add TS-COP as optional approved connector plus biometric search authorization, consent/legal basis, and audit controls.
RFP-P2-02	n-level deep link analysis and kingpin discovery	Y	Present, but computation mode, scoring formula, and performance limits need definition.	Specify async graph jobs, depth limits, ranking logic, and result explanation fields.
RFP-P2-03	Technical analysis report through CDAT including route maps, home/office inference, top contacts, IMEI history, switch-off/silence	Y	Present, but inference rules and unavailable-data behavior need precision.	Add configurable dwell-time rules, ranking logic, silence thresholds, and explicit unavailable-data handling.
RFP-P2-04	LBS tracking/geofencing for Top 50 high-value targets with alerts for Goa/Bengaluru/Mumbai/Orissa Agency	Y	Present, but event latency, geofence source, and escalation behavior are incomplete.	Add watchlist management, geofence master data, alert suppression, event SLA, and escalation rules.
RFP-P2-05	Tower dump analytics with descending criminal/crime link ranking	Y	Present, but ingestion scale, normalization rules, and ranking formula are missing.	Add batch processing rules, number normalization, ranking factors, and async progress tracking.
RFP-P2-06	Drug offender role classification across all NDPS-relevant drug types	Y	Present, but label set, evidence requirements, and fallback rules need definition.	Add controlled role taxonomy, evidence snippets, confidence thresholds, and manual review rules.
RFP-P2-07	Train/tune AI model using Department-provided 50 perfect FIRs	Y	Present, but training governance, benchmark process, and model release controls are missing.	Add model governance, benchmark evaluation, approval workflow, and rollback/manual-only fallback.
RFP-P2-08	Cross-domain data correlation for individuals, vehicles, locations, and events	Y	Present conceptually, but canonical entity model and relationship persistence need more detail.	Add canonical entity/relationship schema and survivorship logic across domains.
RFP-P2-09	Public grievance direct-entry interface	Y	Present, but duplicate detection, SLA, and channel-specific fields need precision.	Add required data fields, duplicate warning rules, SLA priorities, and closure codes.
RFP-P2-10	Automated memo generation and routing to relevant officer with audit trail	Y	Present, but routing rules, approval thresholds, and dispatch audit records need detail.	Add routing matrix, memo statuses, named approver, dispatch channels, and audit artifacts.
RFP-TECH-01	On-prem deployment on private server/local infrastructure	Y	Present, but infrastructure topology and environment sizing are not fully specified.	Add deployment topology, environments, network zones, and baseline hardware sizing.
RFP-TECH-02	Fine-tuned open source LLMs only; no public APIs	Y	Present, but approved model registry, inference boundary, and outbound network controls are not defined.	Add model registry, inference service boundary, egress restrictions, and compliance controls.
RFP-TECH-03	Optimize LLM for intelligent search, summarization, insights, anomaly detection, and natural-language querying	Y	Present via NL query and AI references, but no explicit function boundaries or benchmark criteria exist.	Add AI capability map, model-task mapping, and measurable validation targets.
RFP-TECH-04	DOPAMS software suite shall cover all Phase 1 and Phase 2 modules	Y	BRD phases cover modules, but contractual baseline release acceptance is unclear.	Add release plan, acceptance gates, and dependency-based rollout mapping.
RFP-TECH-05	6-8 months on-site engineering support	Y	Present, but staffing levels, coverage, and support obligations are not measurable.	Add support model, staffing assumptions, ticket categories, response times, and deliverables.
RFP-TECH-06	MIS reports as per Department requirement	Y	Present, but report governance and change request handling are not explicit.	Add report catalog, configurable layout rules, and change management.
RFP-TECH-07	Role-based access system	Y	Present, but permission matrix by action/entity is not fully enumerated.	Add role-action matrix and jurisdiction-aware authorization rules.
RFP-TECH-08	Scalable and customizable as per Department needs	Y	Present, but target scale and admin-configurable elements need precision.	Add scalability targets, configuration domains, and extension rules.
RFP-TECH-09	Bidder technical document to specify stack, AI, features, validations, and security	N	Current BRD references preferred stack but does not define required technical delivery artifacts.	Add implementation guidance section and handover artifact checklist covering stack, security, AI, validation, and infra.
RFP-TECH-10	Solution ownership and customization by EAGLE Force Telangana	Y	Present in assumptions only; IP ownership and artifact handover are not explicit.	Add ownership clause, source-code/model/config handover requirements, and admin rights.
RFP-TECH-11	Bidder shall inform required hosting infrastructure (hardware/software/OS)	N	No formal infrastructure requirement specification is included.	Add infrastructure bill of requirements with compute, GPU, storage, OS, network, and backup prerequisites.
RFP-TECH-12	1-year warranty including maintenance and upgrades	Y	Present, but service levels and scope boundaries are not defined.	Add warranty coverage matrix, response/resolution SLA, and excluded events.

RFP Clause	Clause Summary	Present in Current BRD? (Y/N)	Gap Description	Required Enhancement
RFP-TECH-13	Training to Department officials and 3-4 technical manpower for 2 weeks post deployment	Y	Present, but training outputs, audience, and acceptance criteria are not defined.	Add training plan, materials, attendance, evaluation, and hypercare staffing plan.
RFP-TECH-14	ATS after one year if required	Y	Present, but transition inputs and support handover are not defined.	Add ATS transition checklist and data/config handover requirements.
RFP-CON-01	Deliver solution within 6 weeks of purchase order	N	BRD proposes 3-4 month phases, creating a direct timeline conflict with the tender.	Add contractual delivery baseline, phased deployment interpretation, and clarification dependency.
RFP-CON-02	1 week allowed for installation	N	Installation window is not explicitly captured in the BRD.	Add installation milestone, environment readiness checklist, and acceptance entry criteria.
RFP-CON-03	Payment terms: 80% after delivery/installation, 20% after satisfactory performance after 60 days	N	Commercial milestone dependencies are absent from the BRD.	Add a non-system contractual appendix with payment-linked acceptance events.
RFP-CON-04	Late delivery/installation penalties up to 10%, order cancellation after 30 days	N	BRD does not capture contractual penalty triggers or milestone governance.	Add contractual appendix with delivery milestone control points and risk tracking.
RFP-CON-05	PBG 5% and warranty conditions	N	Warranty is mentioned, but performance guarantee and acceptance dependencies are not.	Add contractual appendix summarizing PBG and warranty obligations for program governance.
RFP-CON-06	No consortium; no subcontracting	N	Procurement constraint is absent from the BRD.	Add procurement applicability note in contractual appendix; mark as non-system design constraint.
RFP-CON-07	Applicable law, jurisdiction, force majeure, notices	N	Contract administration clauses are not represented in the BRD.	Add a contractual governance appendix to preserve non-system RFP obligations.

Appendix B - RFP Traceability Matrix

RFP Clause	Clause Summary	Refined BRD Coverage	Coverage Status
RFP-BIZ-01	Business objective: AI-powered intelligence platform for narcotics trafficking and digital drug promotion networks	1.1, 2.1, FR-19, FR-20	Full
RFP-BIZ-02	Cross-platform monitoring	FR-19, INT-009, INT-013	Full with clarification required
RFP-BIZ-03	AI-based content categorization	FR-20, NFR-AIQ-003, FR-26	Full with clarification required
RFP-BIZ-04	Risk scoring and prioritization	FR-20, WF-ALERT-01, REP-008	Full with clarification required
RFP-BIZ-05	Legal section mapping	FR-21, ENT-CASE.sections_of_law, NFR-AIQ-003	Full with clarification required
RFP-BIZ-06	Digital evidence preservation	FR-22, 5.1, 13, API-14	Full
RFP-BIZ-07	Real-time alerts and escalation	FR-13, FR-24, WF-ALERT-01, ENT-ALERT	Full
RFP-P1-01	54-column full crime history of a person	FR-04, 5.1, SCR-05	Full
RFP-P1-02	Automatic or semi-automatic population of 54-column schema	FR-03, FR-04, FR-25	Full
RFP-P1-03	Phase 1 ingestion from CCTNS, C-DAT, CDR, IPDR, FIR, Confession cum Seizure Memo, IR, C-Trace, eSakshya, Google Drive	FR-02, INT-001..012, ENT-DOC	Full
RFP-P1-04	Phase 3 optional ingestion from ICJS, Transport, e-Prisons, IVFRT, NHAI, Land/Stamp	FR-18, INT-014..020, ENT-CONN	Full
RFP-P1-05	Monthly Report ingestion from Google Drive or similar storage	FR-05, INT-010, REP-001	Full
RFP-P1-06	Extract 20 key MR parameters and update central database	FR-05, 5.3 KPI dictionary, REP-001	Full with clarification required
RFP-P1-07	Automated pictorial and statistical master reports	FR-17, REP-001, REP-002, REP-012	Full
RFP-P1-08	Iterative E-Courts scraping for judgments and bail orders	FR-06, WF-LEGAL-01, ENT-LEGAL	Full
RFP-P1-09	Transaction-CDR cross-check	FR-07, ENT-TXN, API-07	Full
RFP-P1-10	Automatic draft email/request to Unocross for bank statements	FR-07, WF-REQ-01, SCR-12	Full
RFP-P1-11	Automated interrogation report generation	FR-08, API-06, REP-004	Full
RFP-P1-12	Mandatory fixed templates for interrogation report	FR-08, FR-23, SCR-15	Full
RFP-P1-13	Panchanama ingestion in text/image/OCR Telugu and English with missing-field manual entry	FR-03, FR-08, SCR-04, NFR-LOC-001	Full
RFP-P2-01	One-click dossier by name/alias/surname/phone/biometric against CCTNS/TS-COP	FR-09, INT-013, SCR-06	Full with clarification required
RFP-P2-02	n-level deep link analysis and kingpin discovery	FR-11, WF-NET-implicit, REP-006	Full
RFP-P2-03	Technical analysis report through CDAT including route maps, home/office inference, top contacts, IMEI history,	FR-12, ENT-COMM, ENT-LOC, REP-005	Full

RFP Clause	Clause Summary	Refined BRD Coverage	Coverage Status
	switch-off/silence		
RFP-P2-04	LBS tracking/geofencing for Top 50 high-value targets with alerts for Goa/Bengaluru/Mumbai/Orissa Agency	FR-13, ENT-ALERT, SCR-10	Full
RFP-P2-05	Tower dump analytics with descending criminal/crime link ranking	FR-14, REP-006, API-07	Full
RFP-P2-06	Drug offender role classification across all NDPS-relevant drug types	FR-15, ENT-SUBJ.offender_role	Full
RFP-P2-07	Train/tune AI model using Department-provided 50 perfect FIRs	FR-15, FR-26, NFR-AIQ-002	Full
RFP-P2-08	Cross-domain data correlation for individuals, vehicles, locations, and events	FR-04, FR-09, FR-25, ENT-NODE/EDGE	Full
RFP-P2-09	Public grievance direct-entry interface	FR-16, SCR-11, ENT-LEAD	Full
RFP-P2-10	Automated memo generation and routing to relevant officer with audit trail	FR-16, FR-24, WF-LEAD-01, ENT-MEMO	Full
RFP-TECH-01	On-prem deployment on private server/local infrastructure	2.4, 8, 9, 16.1	Full
RFP-TECH-02	Fine-tuned open source LLMs only; no public APIs	2.4, FR-10, FR-26, NFR-SEC-005	Full
RFP-TECH-03	Optimize LLM for intelligent search, summarization, insights, anomaly detection, and natural-language querying	FR-10, FR-20, FR-21, FR-26, 15.6	Full
RFP-TECH-04	DOPAMS software suite shall cover all Phase 1 and Phase 2 modules	2.1, 10, 16.1	Full
RFP-TECH-05	6-8 months on-site engineering support	16.3, NFR-OBS-001, CQ-17	Full with clarification required
RFP-TECH-06	MIS reports as per Department requirement	FR-17, Section 12	Full
RFP-TECH-07	Role-based access system	FR-01, Section 3, NFR-SEC-003	Full
RFP-TECH-08	Scalable and customizable as per Department needs	2.4, 8, FR-23, 15.1	Full
RFP-TECH-09	Bidder technical document to specify stack, AI, features, validations, and security	15.1-15.6, 16.4	Full
RFP-TECH-10	Solution ownership and customization by EAGLE Force Telangana	2.4, 16.4, CMP-010	Full
RFP-TECH-11	Bidder shall inform required hosting infrastructure (hardware/software/OS)	16.2, 15.2, CQ-14	Full with clarification required
RFP-TECH-12	1-year warranty including maintenance and upgrades	16.3, NFR-REL-001	Full
RFP-TECH-13	Training to Department officials and 3-4 technical manpower for 2 weeks post deployment	16.3, 16.5, CQ-17	Full
RFP-TECH-14	ATS after one year if required	16.3, 16.5	Full
RFP-CON-01	Deliver solution within 6 weeks of purchase order	16.1, Appendix B, CQ-01	Full with clarification required
RFP-CON-02	1 week allowed for installation	16.1, 16.2	Full with clarification required
RFP-CON-03	Payment terms: 80% after delivery/installation, 20% after satisfactory performance after 60 days	16.6, Appendix B	Referenced - contractual/non-system
RFP-CON-04	Late delivery/installation penalties up to 10%, order cancellation after 30 days	16.6, Appendix B	Referenced - contractual/non-system
RFP-CON-05	PBG 5% and warranty conditions	16.3, 16.6	Referenced - contractual/non-system
RFP-CON-06	No consortium; no subcontracting	16.6, Appendix B	Referenced - contractual/non-system
RFP-CON-07	Applicable law, jurisdiction, force majeure, notices	16.6, Appendix B	Referenced - contractual/non-system

Appendix C - Clarification Questions

Question ID	Clarification Question
CQ-01	The tender requires delivery within 6 weeks plus 1 week installation, but the BRD assumes multi-month Phase 1 and Phase 2 delivery. Which interpretation governs contractual acceptance: core MVP in 6 weeks, or full Phase 1+2 rollout?
CQ-02	Please confirm the final Department-approved 54-column dictionary, including whether gang associates should remain fixed 11 columns in outputs or be physically normalized in the database.
CQ-03	Please provide the final list of 20 Monthly Report KPIs, including definitions, formulas, allowed ranges, and reporting month conventions.
CQ-04	Please confirm which Phase 1 sources are available on day 1, their integration method (API/DB/file), and whether sample data plus credentials can be provided for each.
CQ-05	Is TS-COP integration formally approved for face-match search? If yes, what are the legal, process, and audit constraints for biometric use?
CQ-06	For cross-platform monitoring, which channels/platforms are explicitly approved, and does scope include only public OSINT plus lawfully obtained exports, or additional sources?
CQ-07	Please confirm the approved taxonomy and factor weights for AI content categorization and risk scoring, or whether the proposed defaults should be treated as the baseline for UAT.

Question ID	Clarification Question
CQ-08	Please confirm the legal section mapping scope: NDPS only, or NDPS plus IPC/CrPC/IT Act or other allied sections.
CQ-09	Please confirm retention periods for evidence, audit logs, alerts, leads, reports, and model-governance artifacts, especially where legal hold may apply.
CQ-10	Should approved Unocross outputs be exported only, or should the system also send them through Department SMTP automatically after approval?
CQ-11	Please provide geofence coordinates/polygons and the exact meaning of 'Orissa Agency areas' for operational alerting.
CQ-12	Please confirm SLA expectations for critical, high, medium, and low alerts, lead routing, legal review, and approval tasks.
CQ-13	Will Google Drive remain approved for MR sync in production, or must it be replaced with an on-prem file share or other Department-approved bridge?
CQ-14	Please confirm baseline infrastructure: CPU, GPU, RAM, storage, OS, DB/search/object-storage standards, and whether Kubernetes is permitted.
CQ-15	What are the acceptance criteria for the 50 ideal FIR benchmark set, and who is the named approver for extraction, classification, and legal mapping model readiness?
CQ-16	Please confirm whether DOCX export is allowed for final interrogation reports or only for drafts, with final records restricted to PDF.
CQ-17	Please confirm the exact support model for the 6-8 month on-site engineering support period, including working hours, staff count, and ticket SLAs.
CQ-18	Please confirm whether procurement-only clauses such as payment milestones, PBG, and no-subcontracting should be tracked inside the BRD appendix or separately in the project charter/contract register.